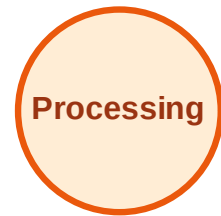


BFS Traversal

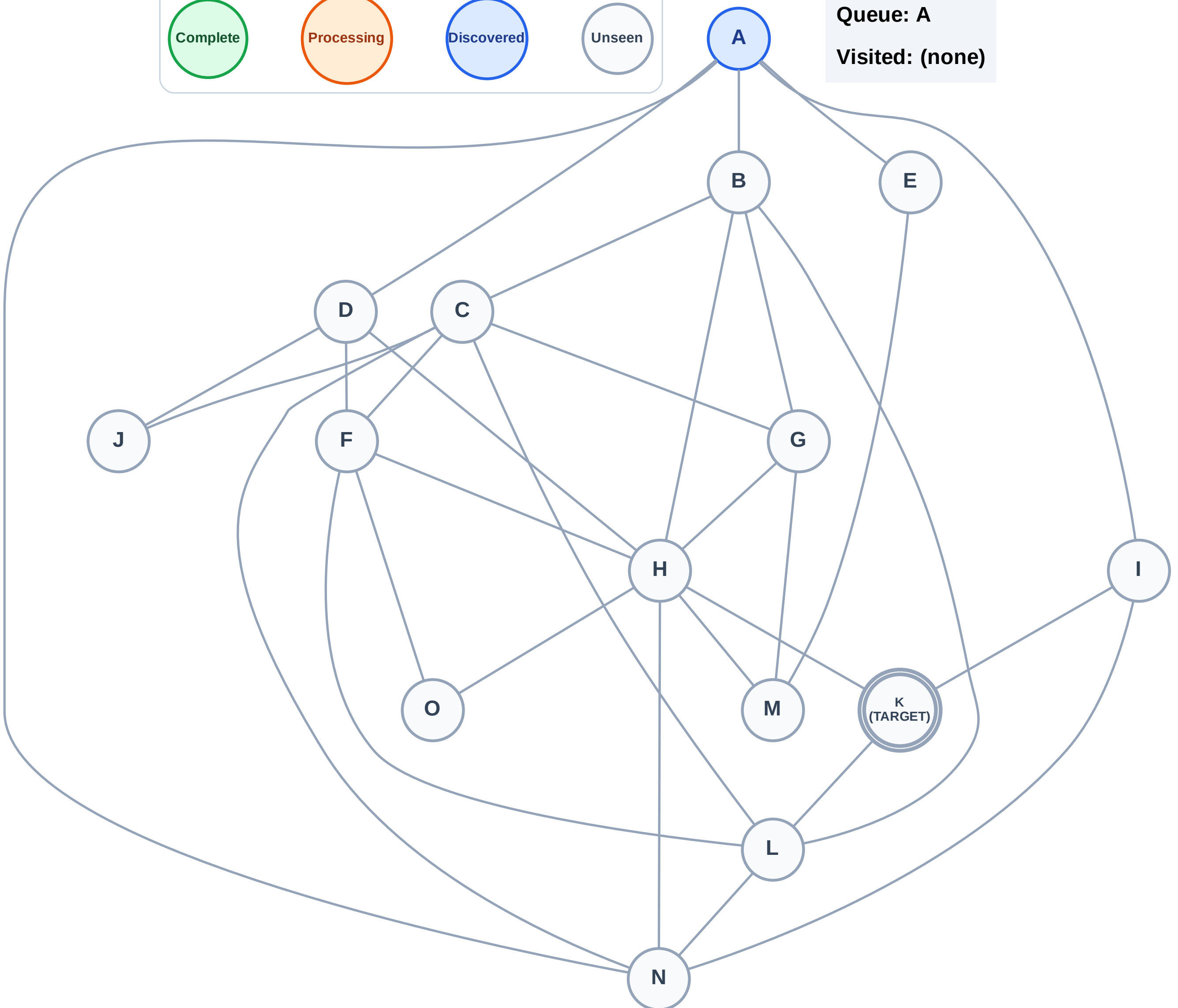
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



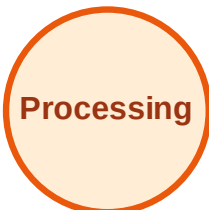
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



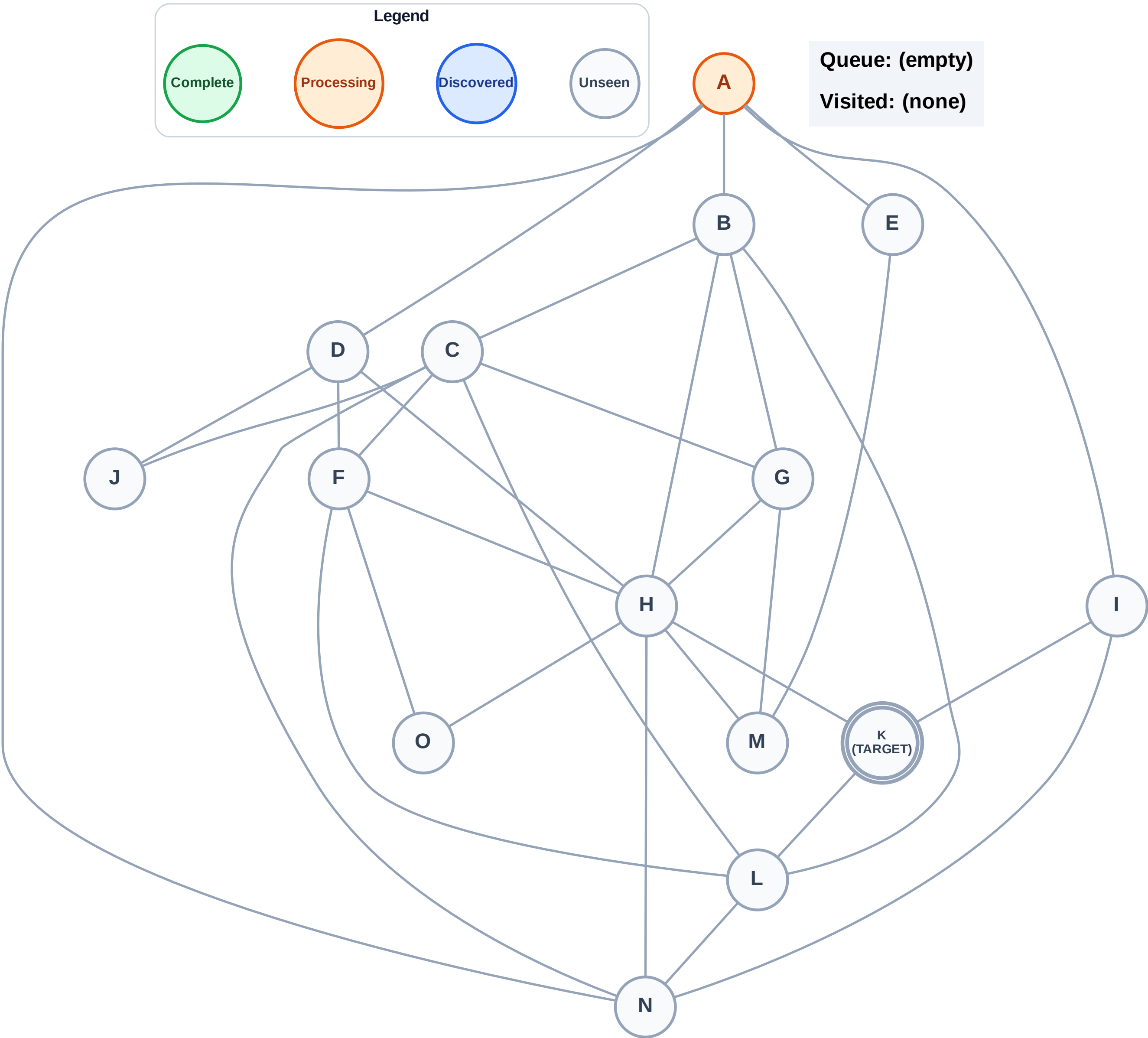
Discovered



Unseen

Queue: (empty)

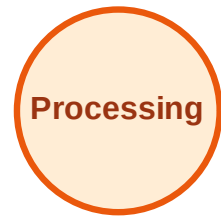
Visited: (none)



BFS Traversal

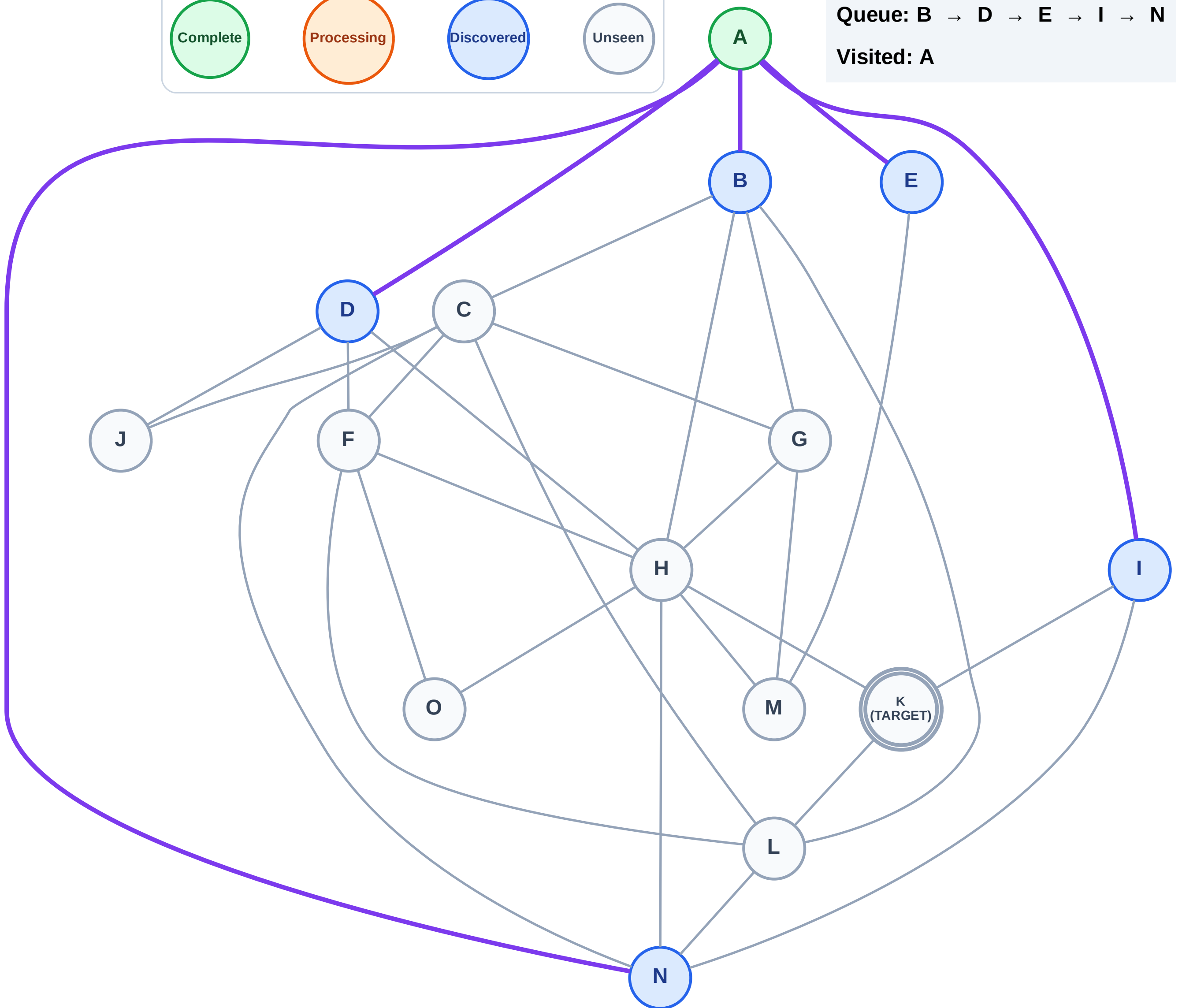
Step 3: Discover B, D, E, I, N from A

Legend



Queue: B → D → E → I → N

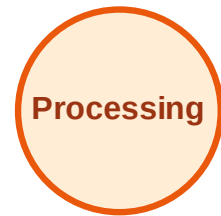
Visited: A



BFS Traversal

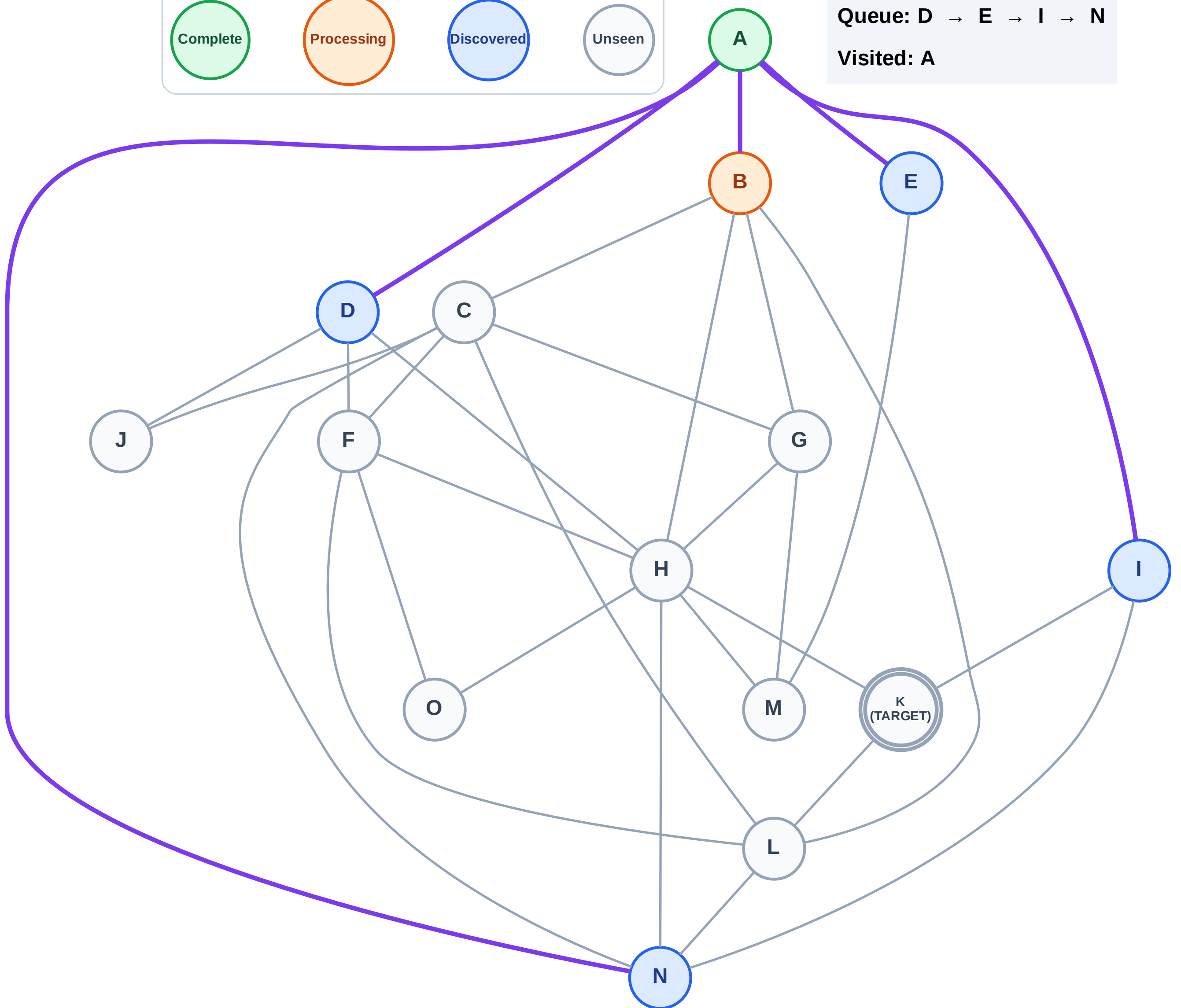
Step 4: Process node B

Legend



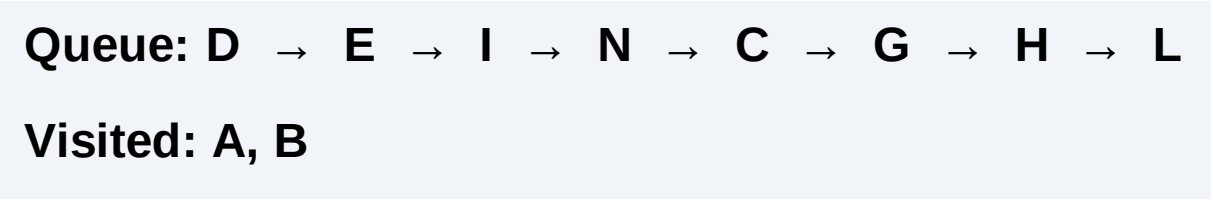
Queue: D → E → I → N

Visited: A



Step 5: Discover C, G, H, L from B

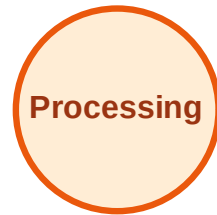
Step 5: Discover C, G, H, L from B



BFS Traversal

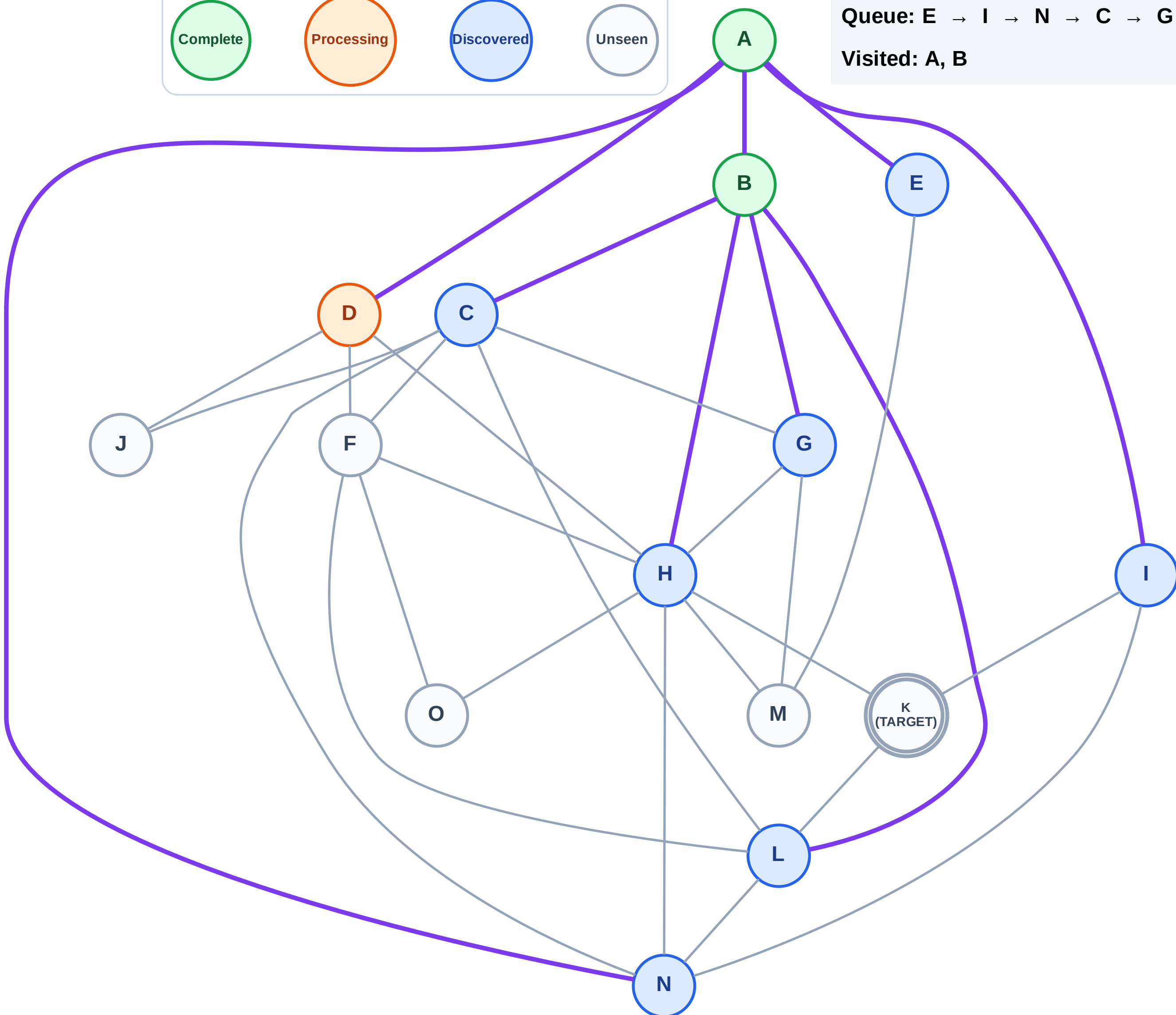
Step 6: Process node D

Legend



Queue: E → I → N → C → G → H → L

Visited: A, B



BFS Traversal

Step 7: Discover F, J from D

Legend

Complete

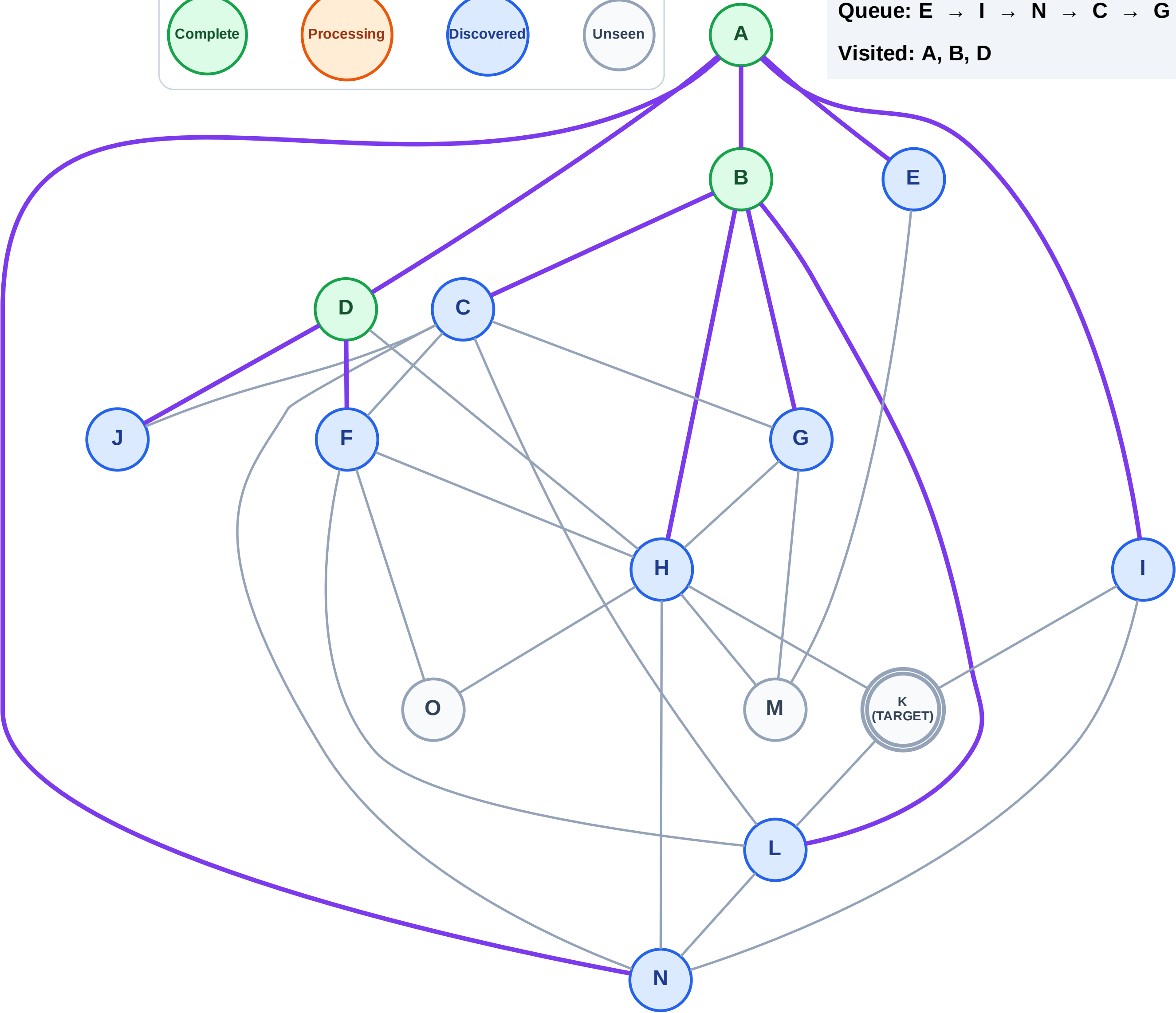
Processing

Discovered

Unseen

Queue: E → I → N → C → G → H → L → F → J

Visited: A, B, D



BFS Traversal

Step 8: Process node E

Legend

Complete

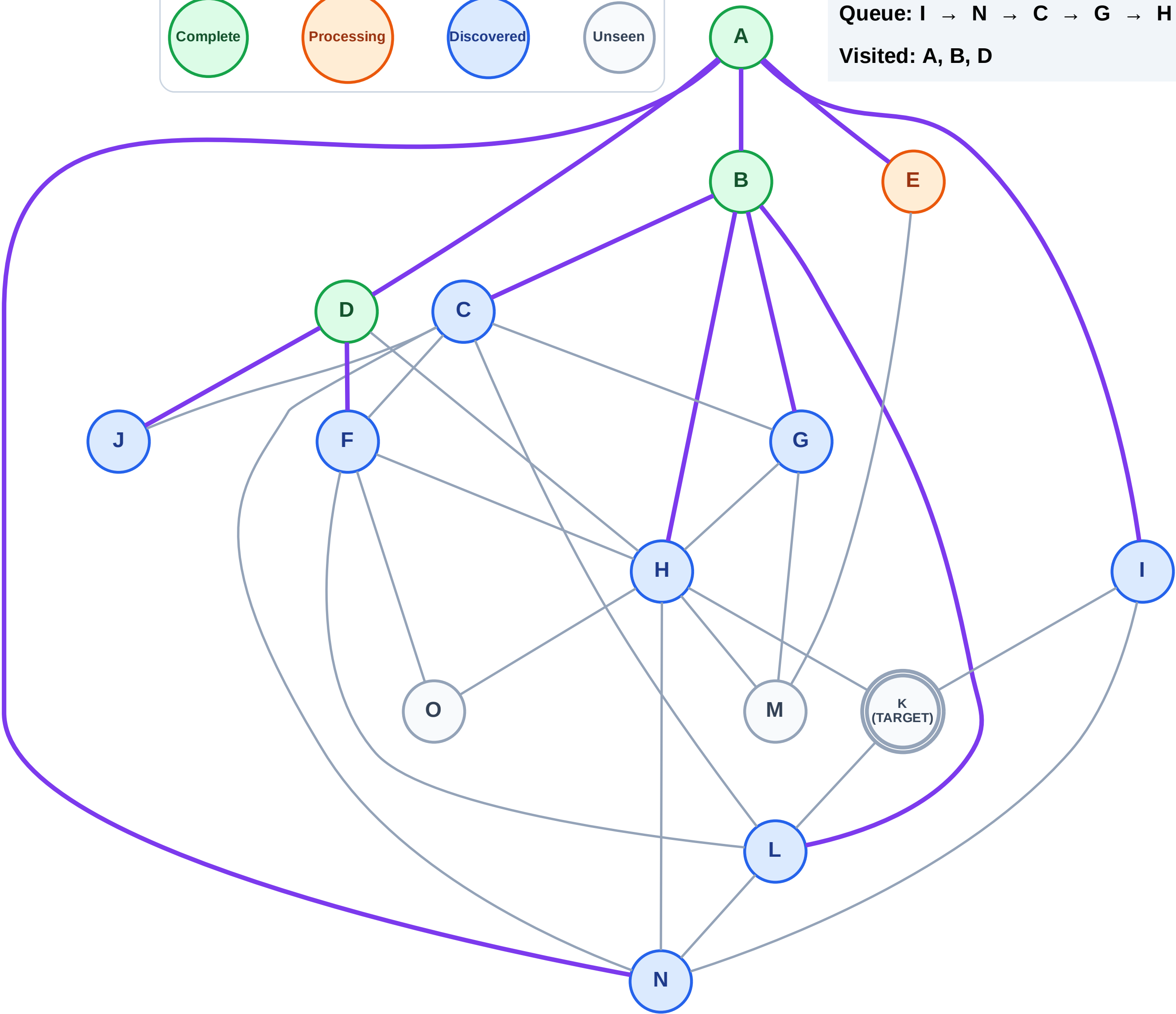
Processing

Discovered

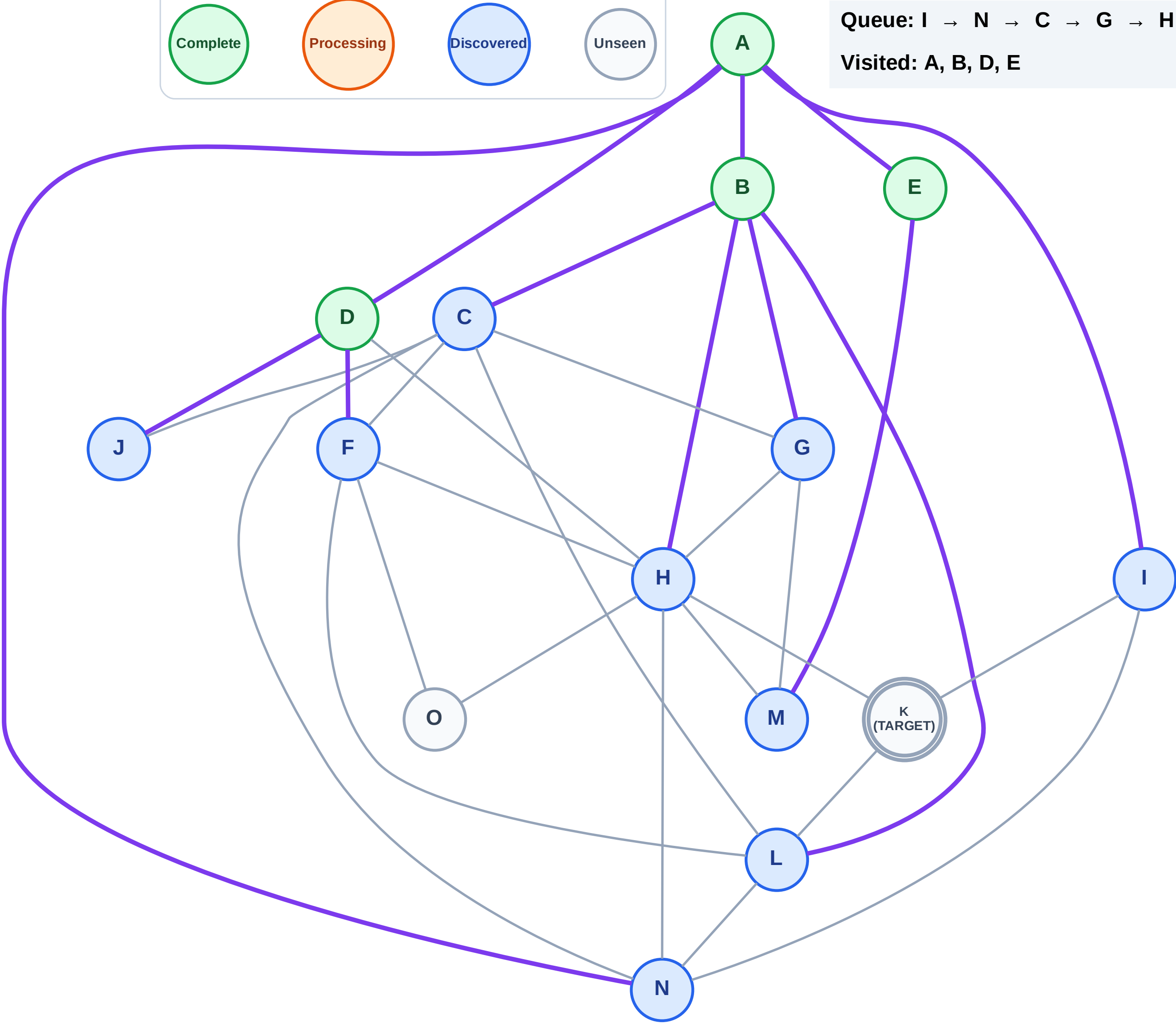
Unseen

Queue: I → N → C → G → H → L → F → J

Visited: A, B, D



Step 9: Discover M from E



BFS Traversal

Step 10: Process node I

Legend

Complete

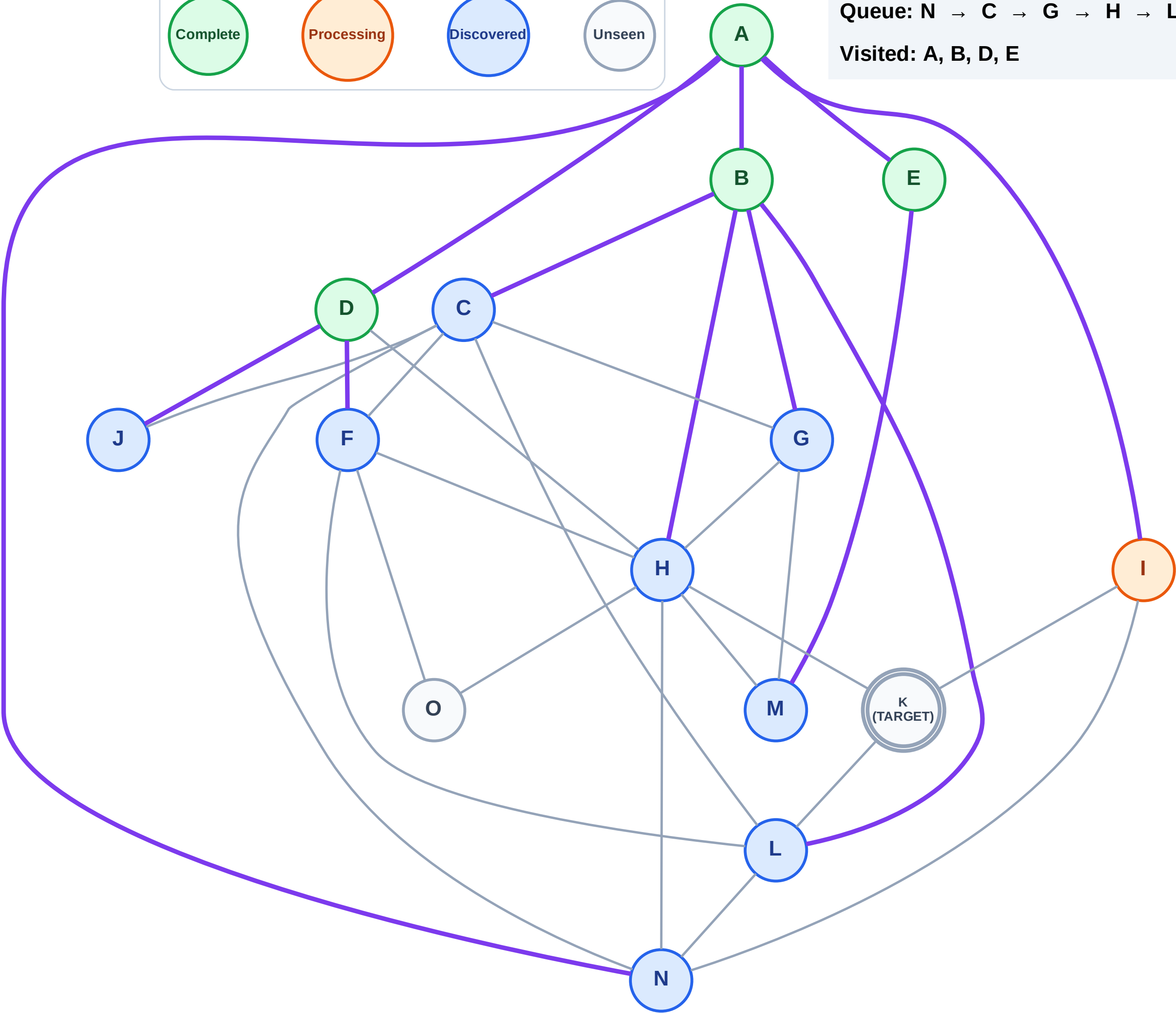
Processing

Discovered

Unseen

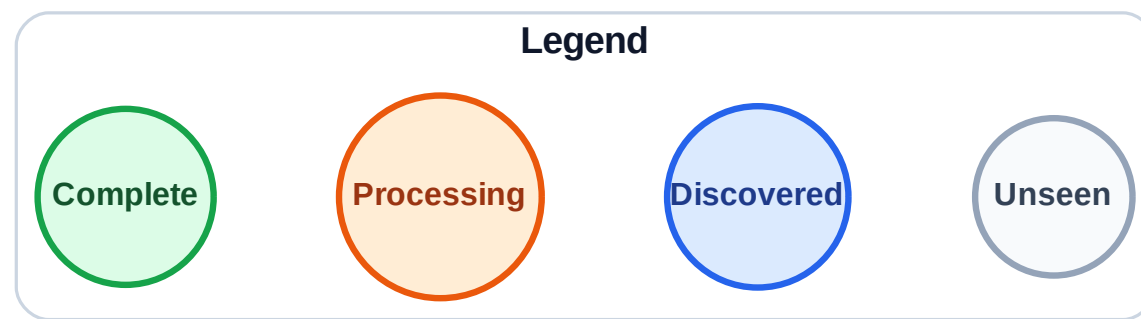
Queue: N → C → G → H → L → F → J → M

Visited: A, B, D, E



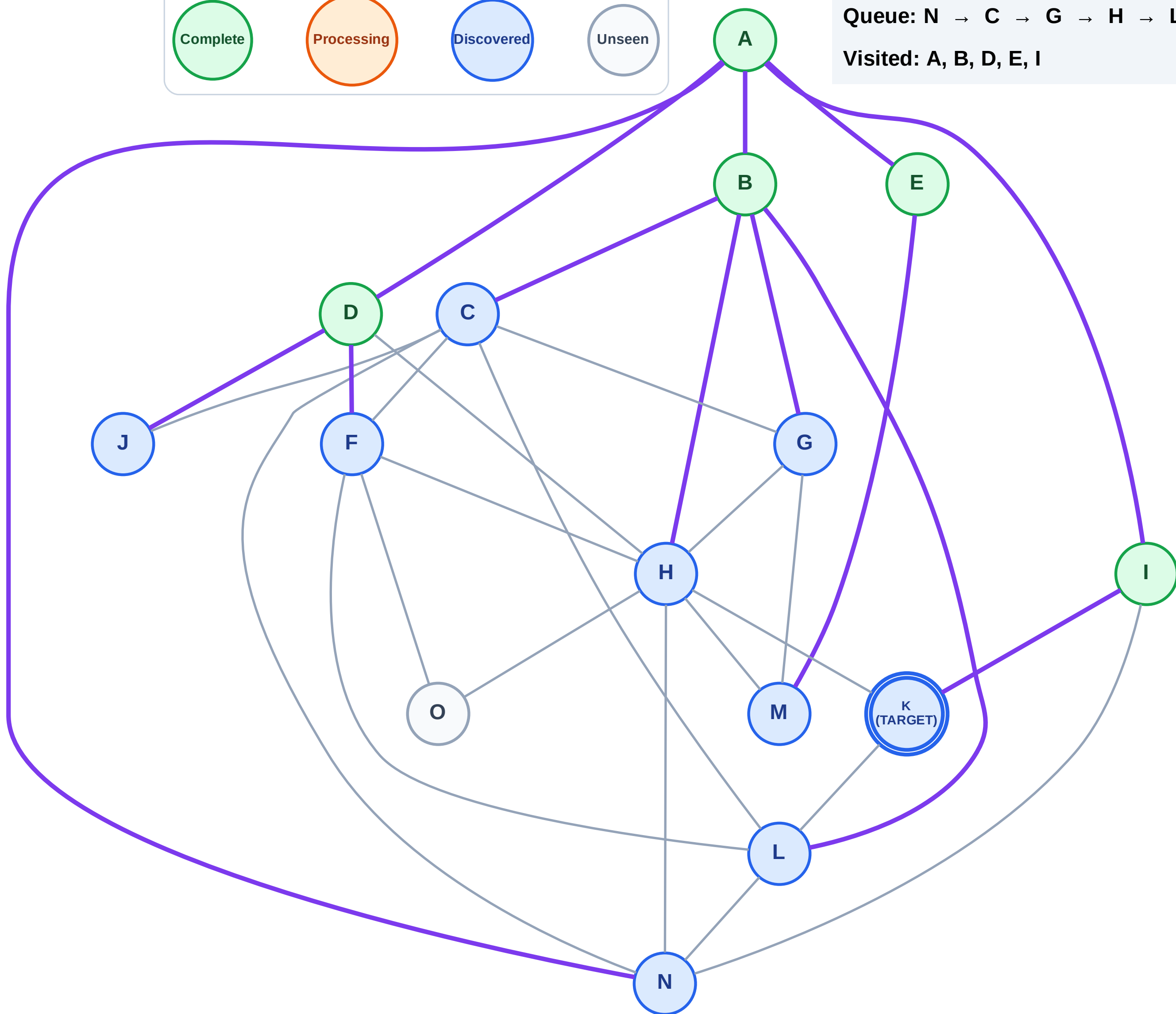
BFS Traversal

Step 11: Discover K from I



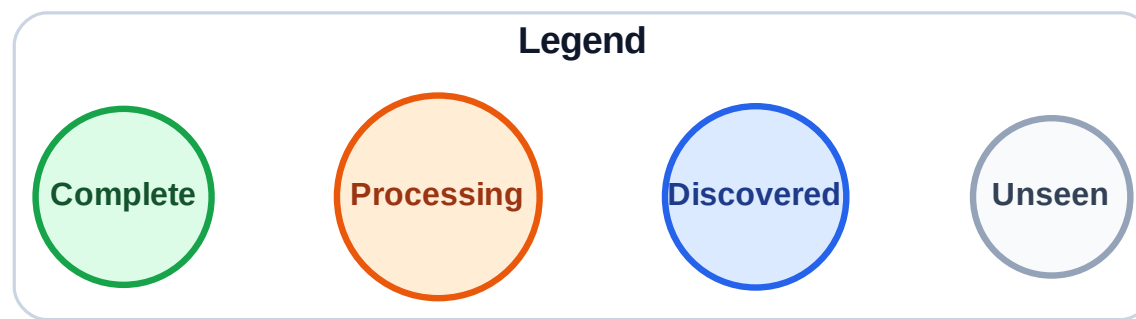
Queue: N → C → G → H → L → F → J → M → K

Visited: A, B, D, E, I

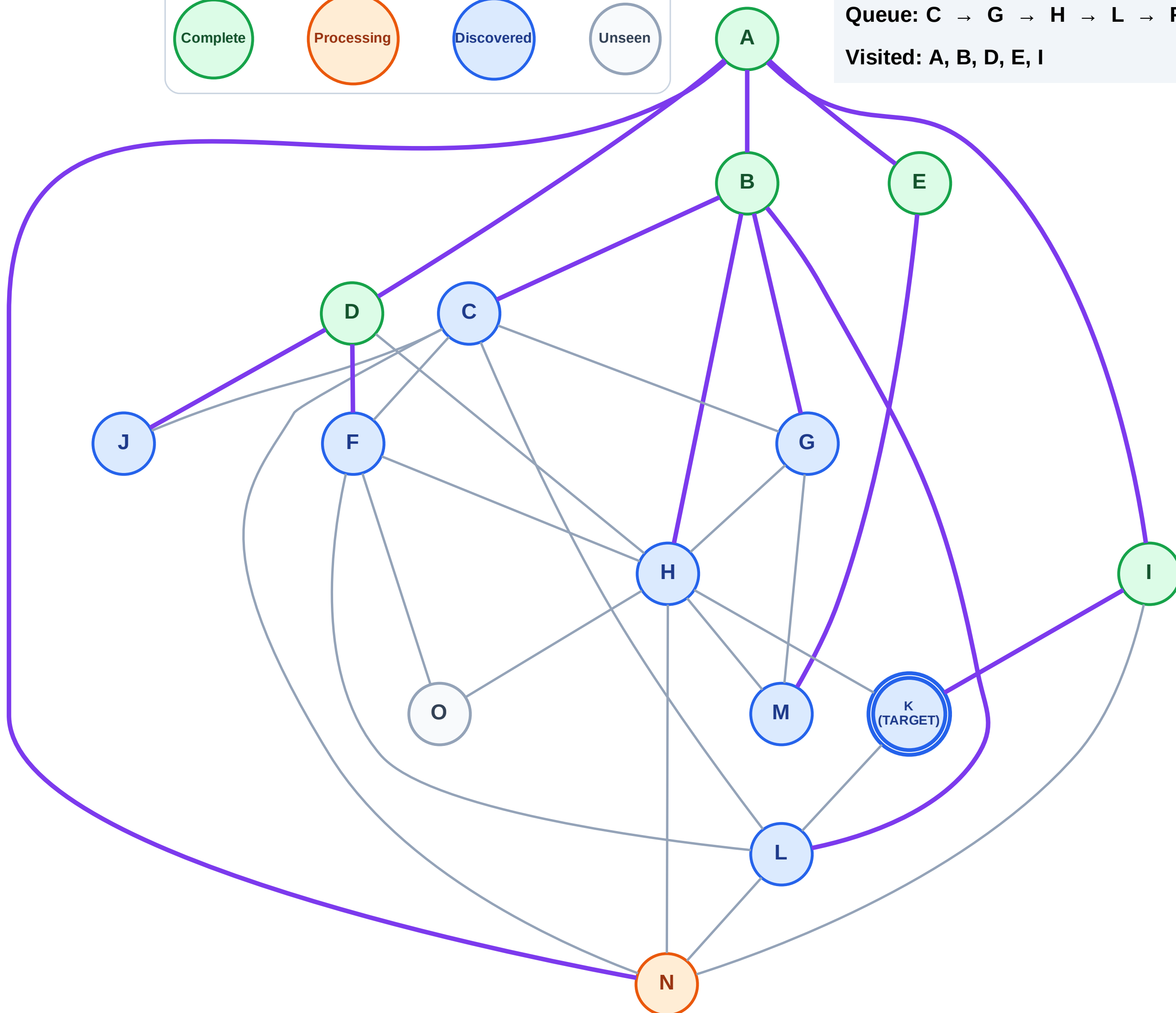


BFS Traversal

Step 12: Process node N



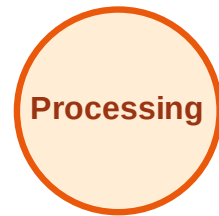
Queue: C → G → H → L → F → J → M → K
Visited: A, B, D, E, I



BFS Traversal

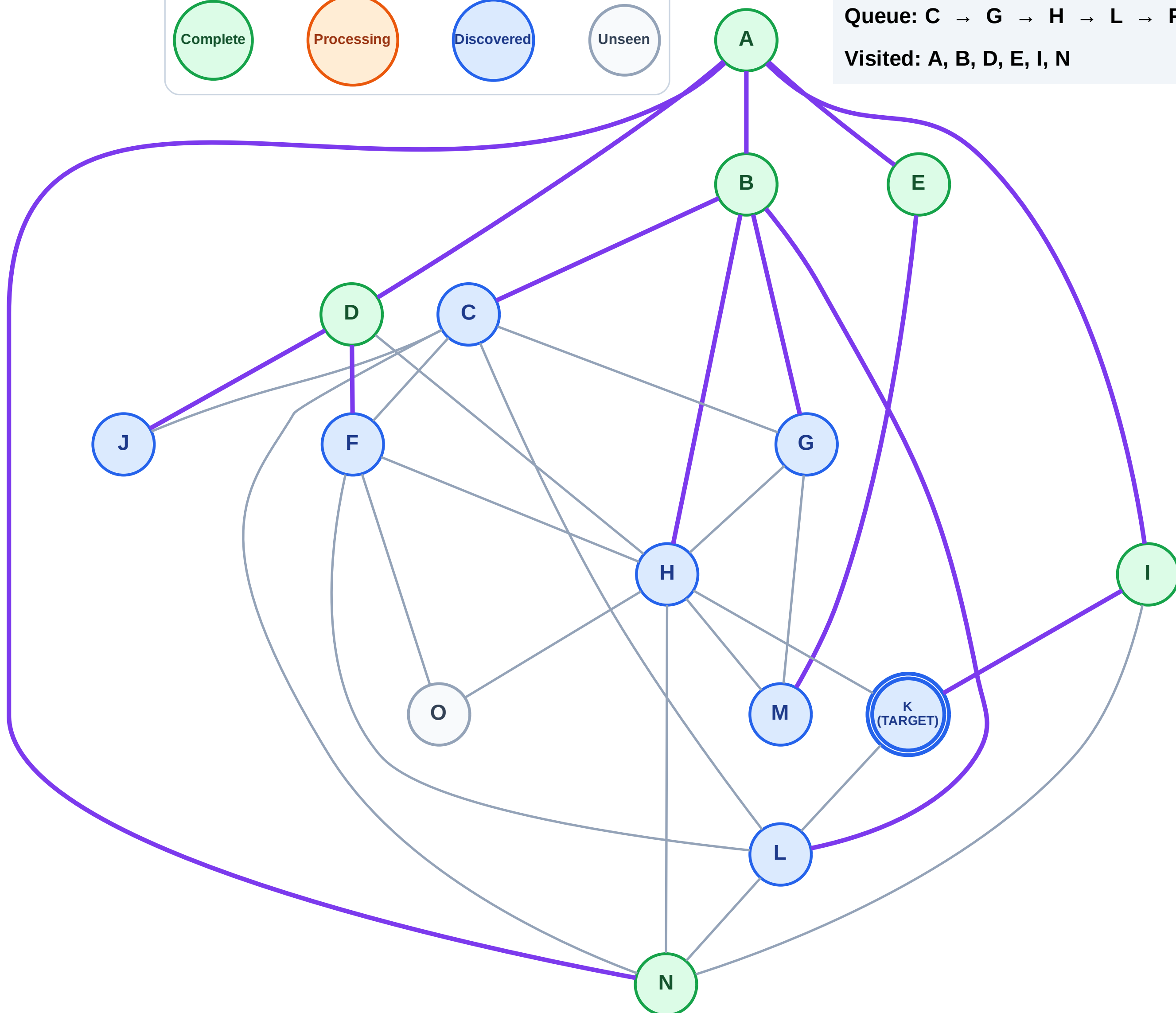
Step 13: No new neighbors from N

Legend



Queue: C → G → H → L → F → J → M → K

Visited: A, B, D, E, I, N



BFS Traversal

Step 14: Process node C

Legend

Complete

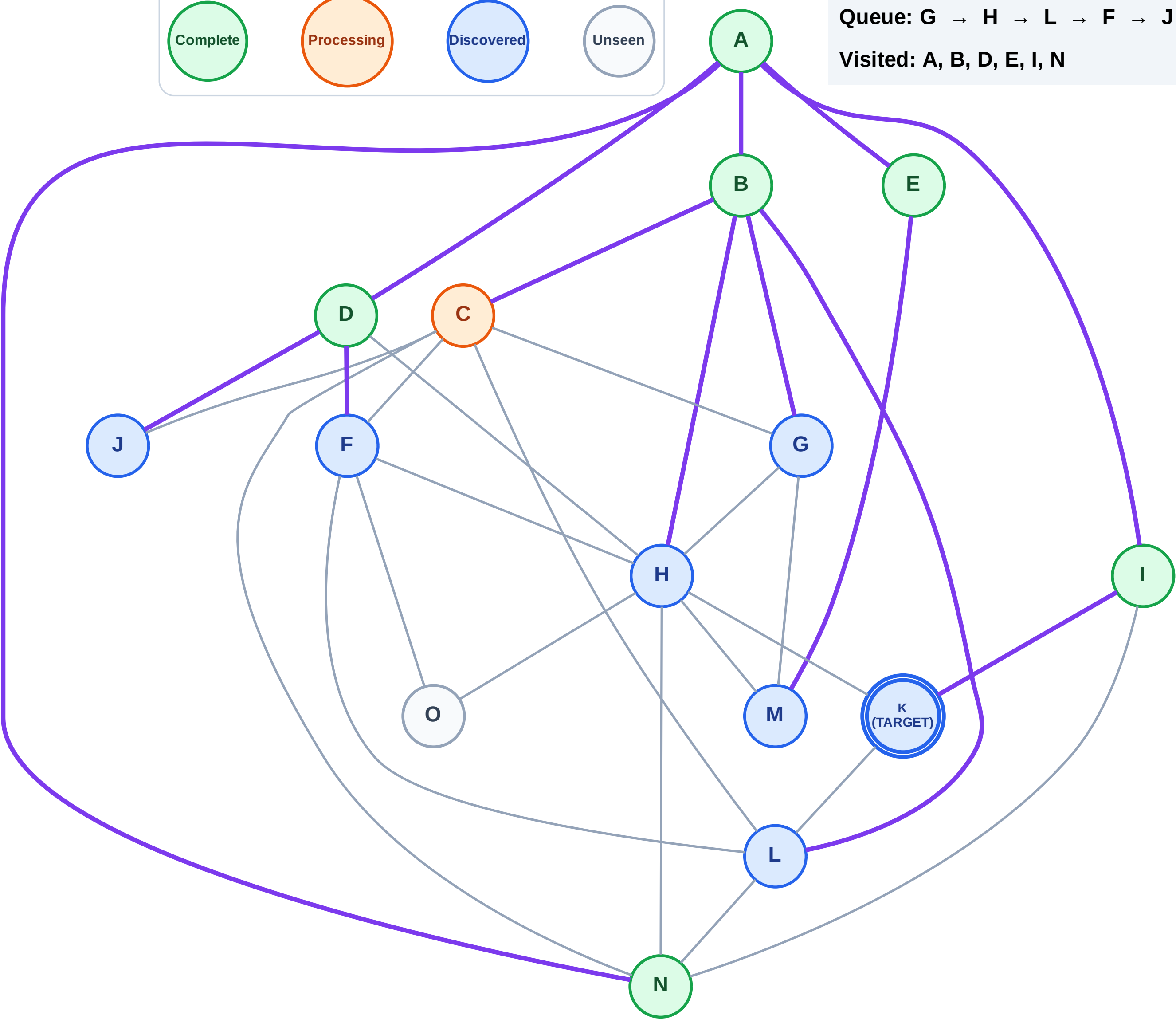
Processing

Discovered

Unseen

Queue: G → H → L → F → J → M → K

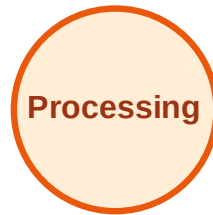
Visited: A, B, D, E, I, N



BFS Traversal

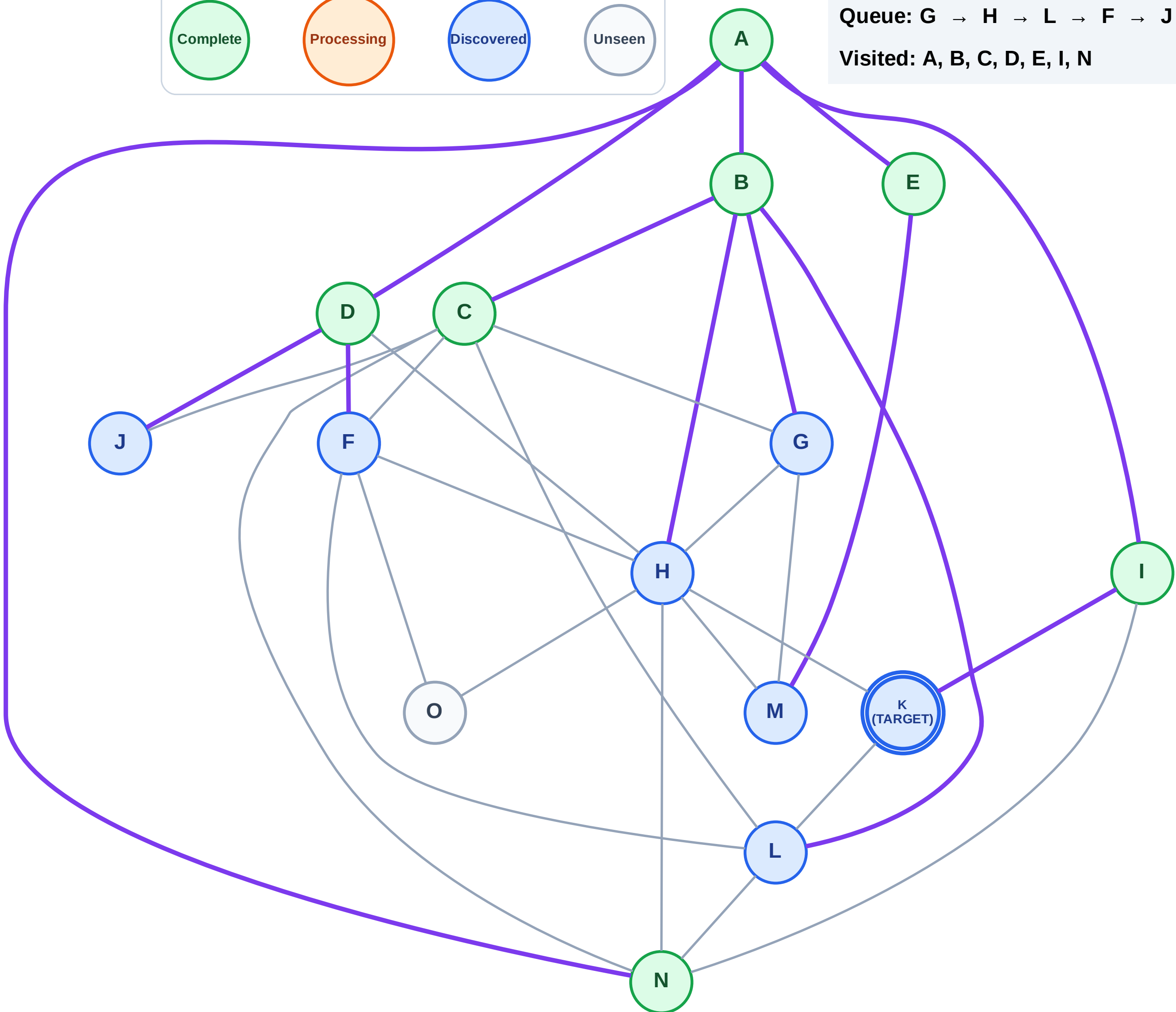
Step 15: No new neighbors from C

Legend



Queue: G → H → L → F → J → M → K

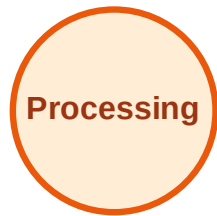
Visited: A, B, C, D, E, I, N



BFS Traversal

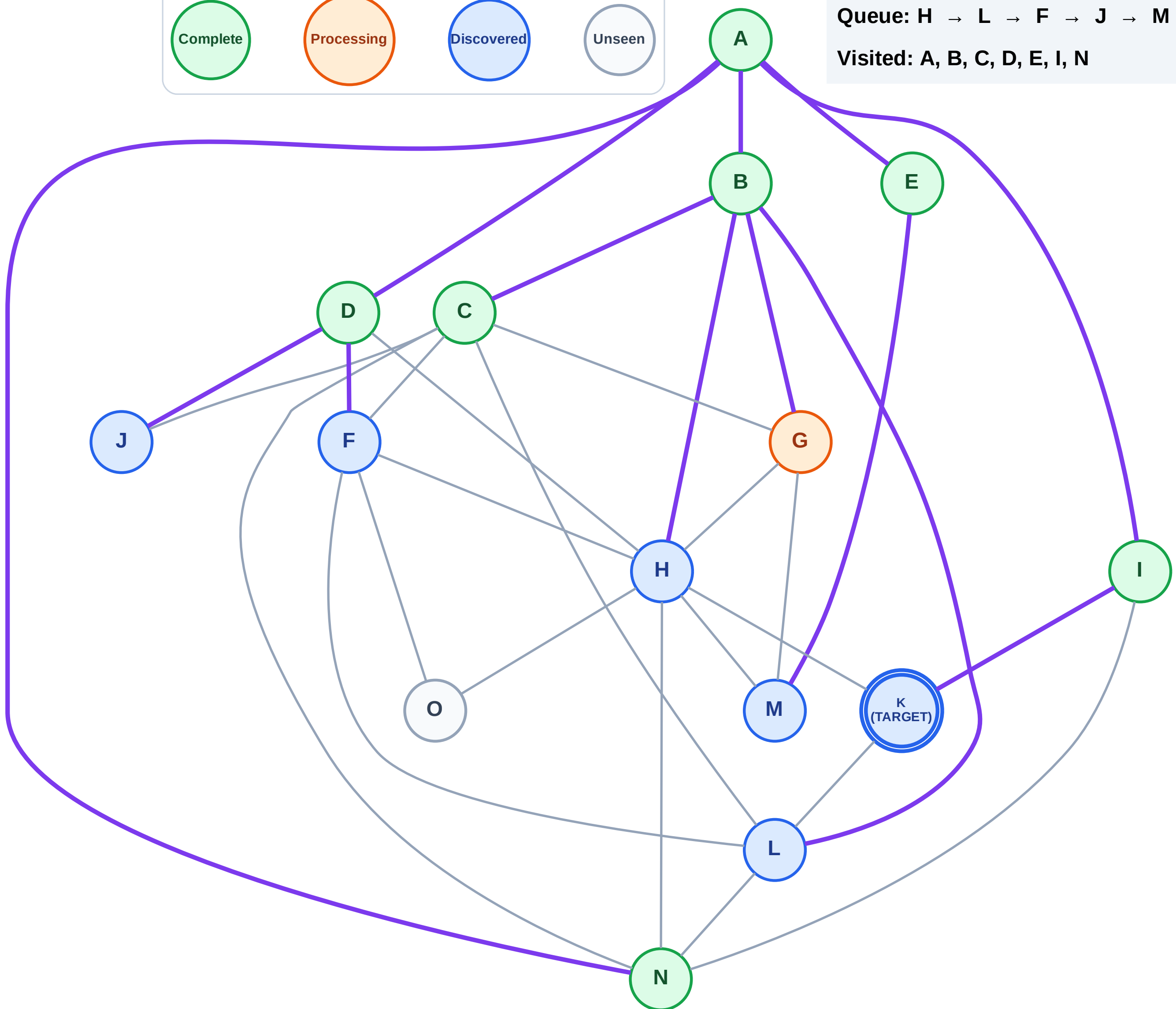
Step 16: Process node G

Legend



Queue: H → L → F → J → M → K

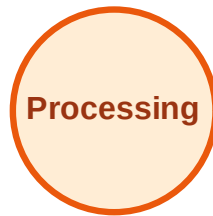
Visited: A, B, C, D, E, I, N



BFS Traversal

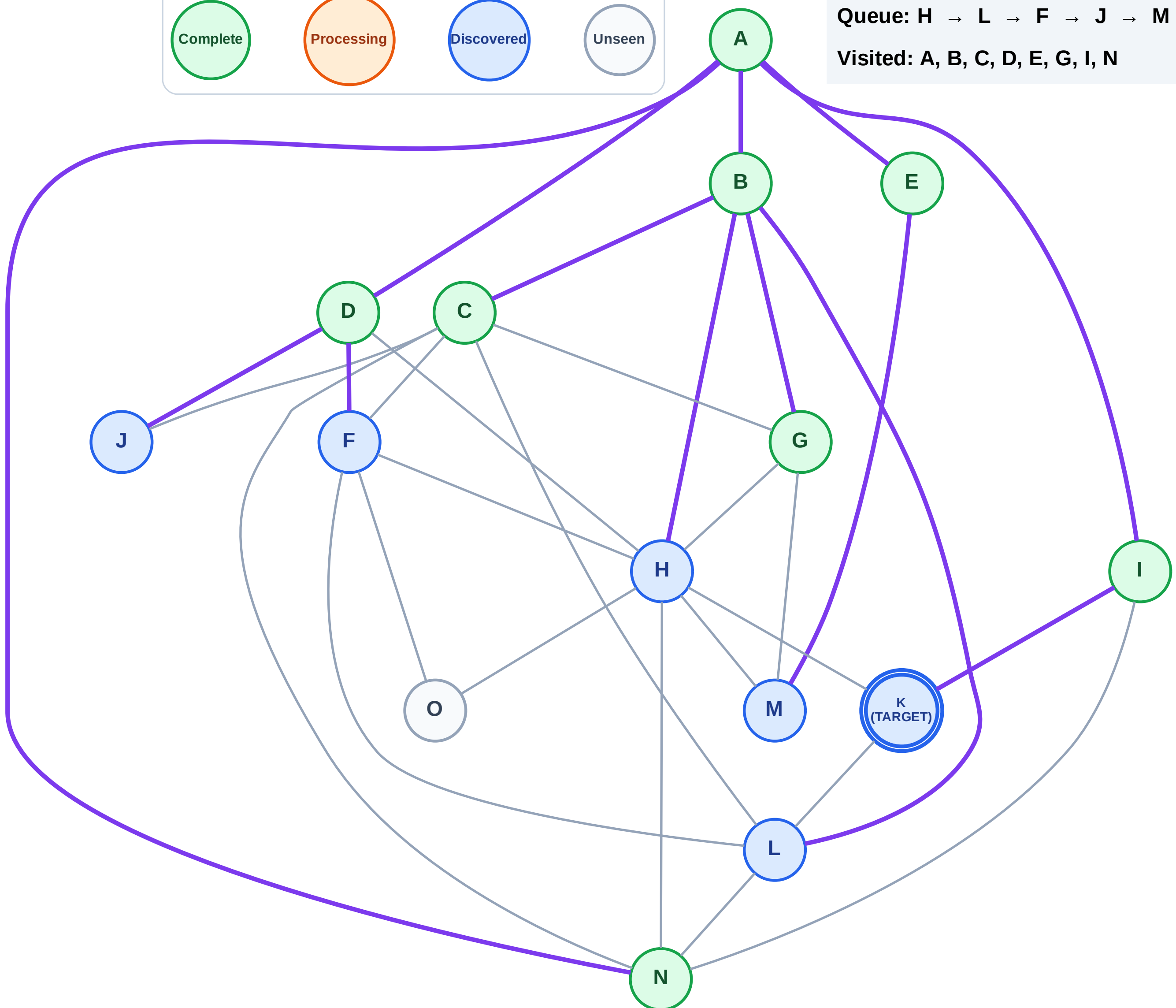
Step 17: No new neighbors from G

Legend



Queue: H → L → F → J → M → K

Visited: A, B, C, D, E, G, I, N



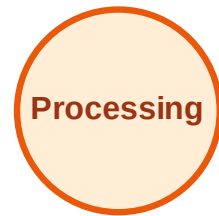
BFS Traversal

Step 18: Process node H

Legend



Complete



Processing



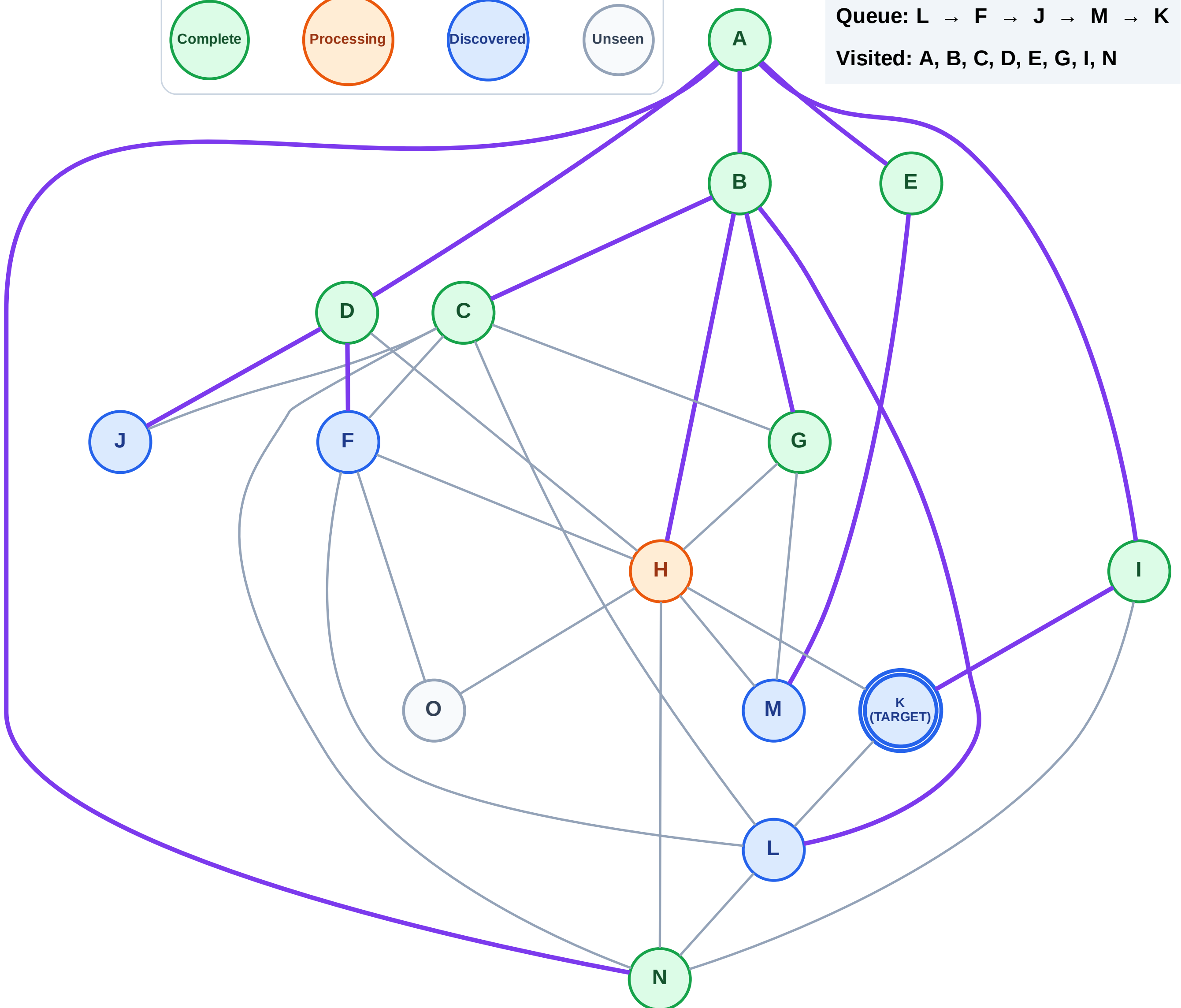
Discovered



Unseen

Queue: L → F → J → M → K

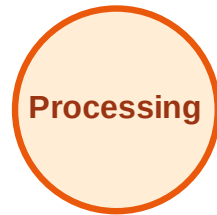
Visited: A, B, C, D, E, G, I, N



BFS Traversal

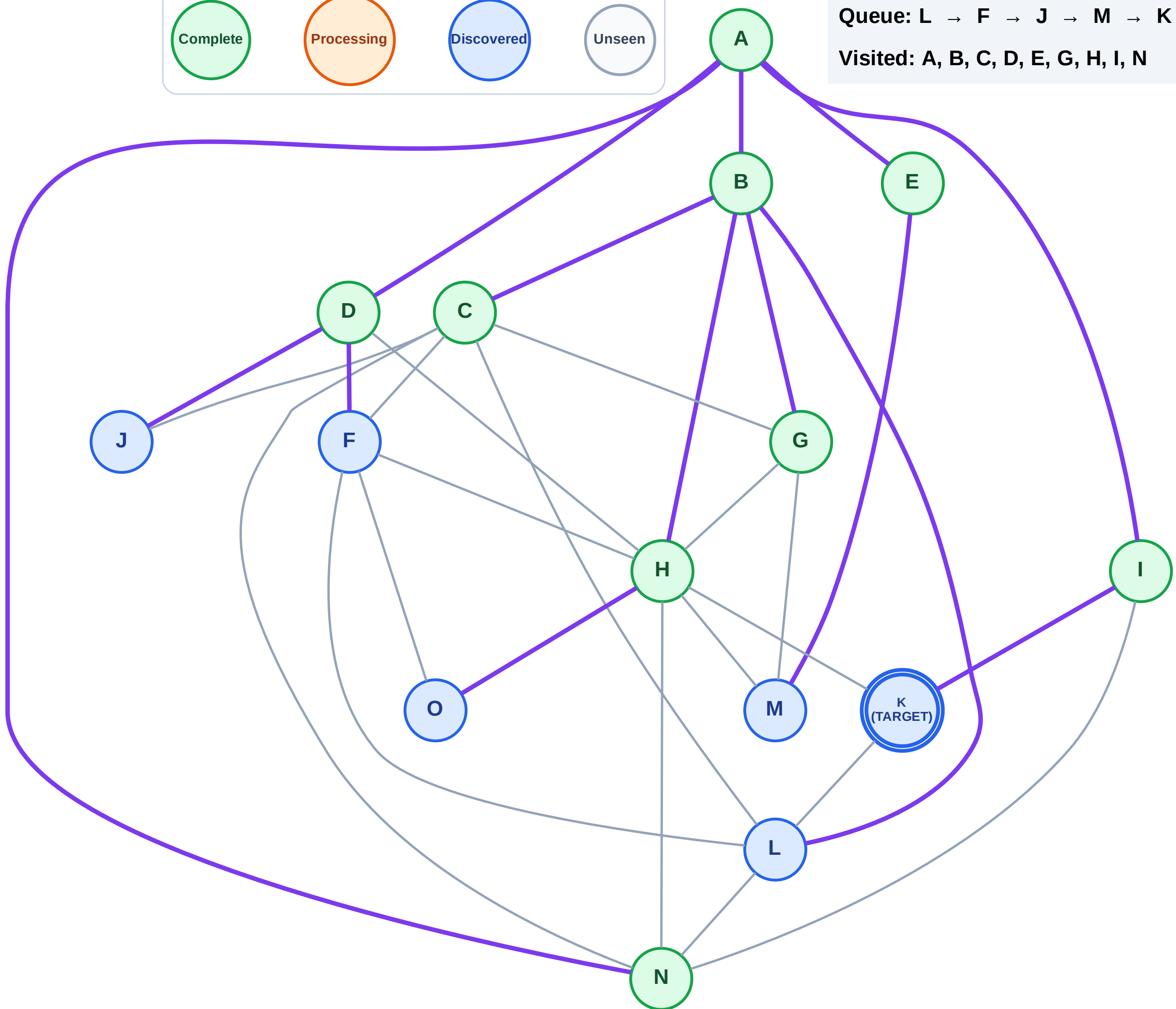
Step 19: Discover O from H

Legend



Queue: L → F → J → M → K → O

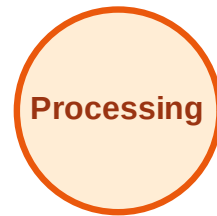
Visited: A, B, C, D, E, G, H, I, N



BFS Traversal

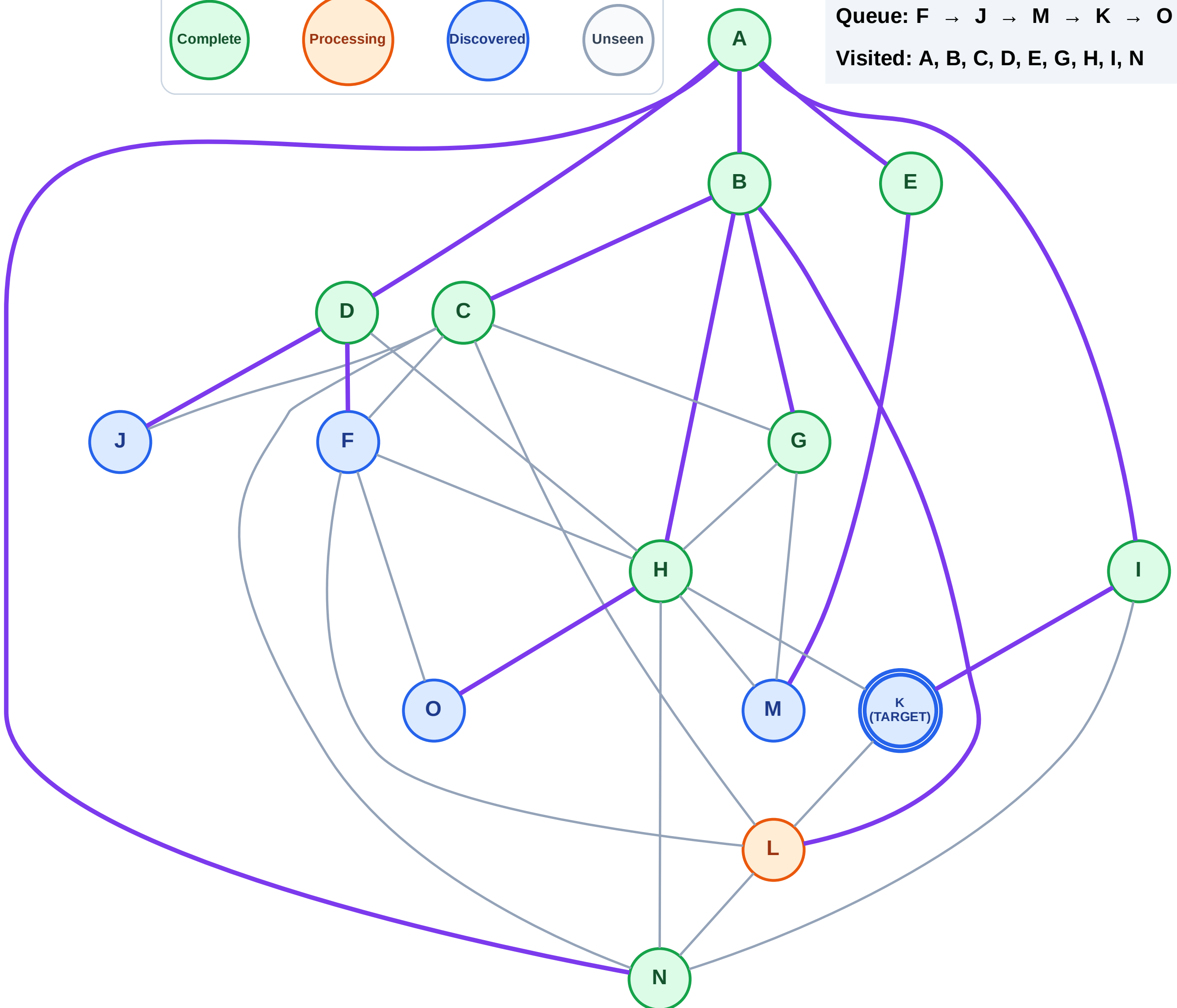
Step 20: Process node L

Legend



Queue: F → J → M → K → O

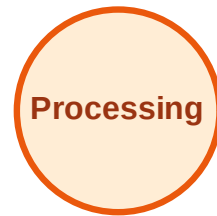
Visited: A, B, C, D, E, G, H, I, N



BFS Traversal

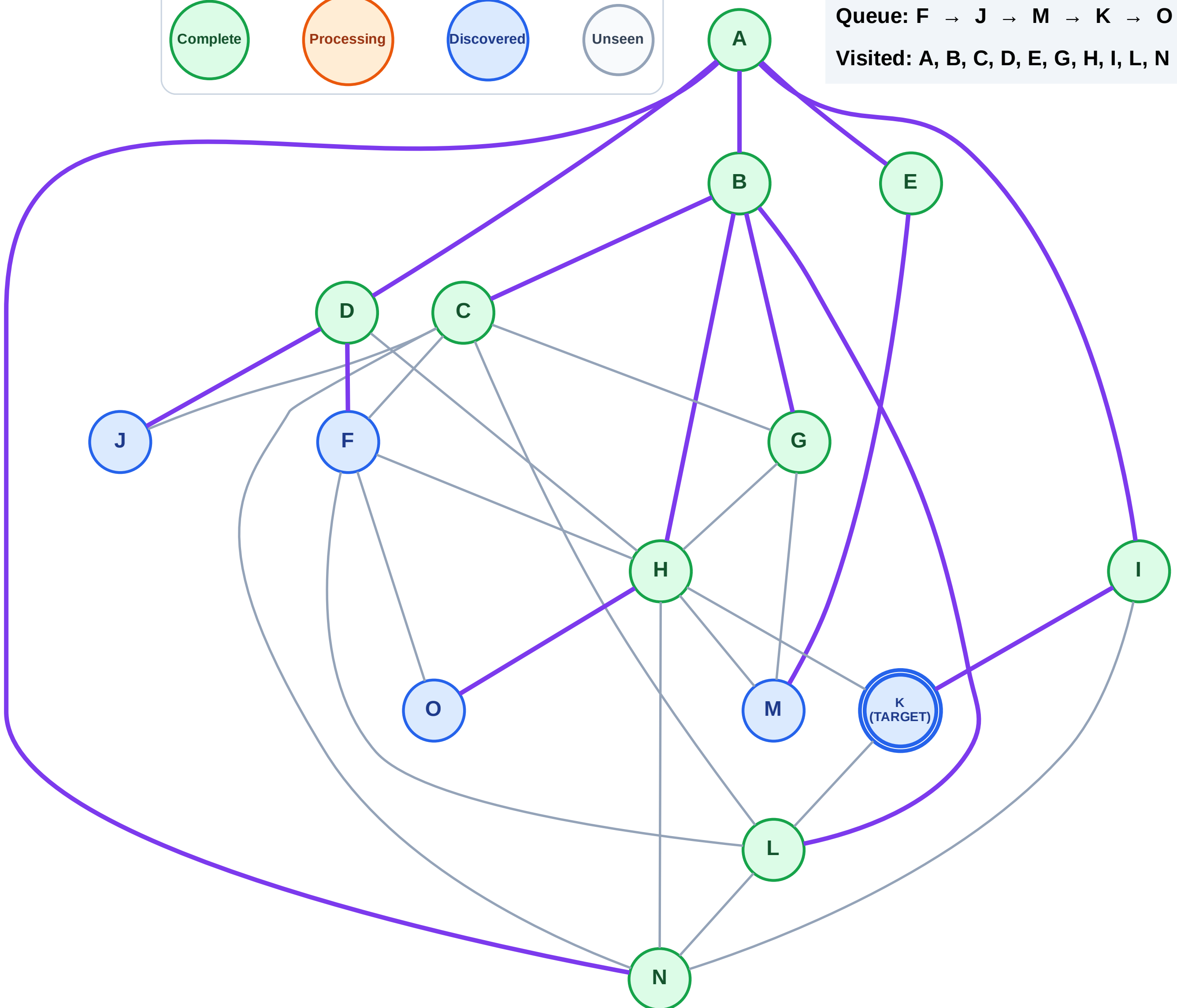
Step 21: No new neighbors from L

Legend

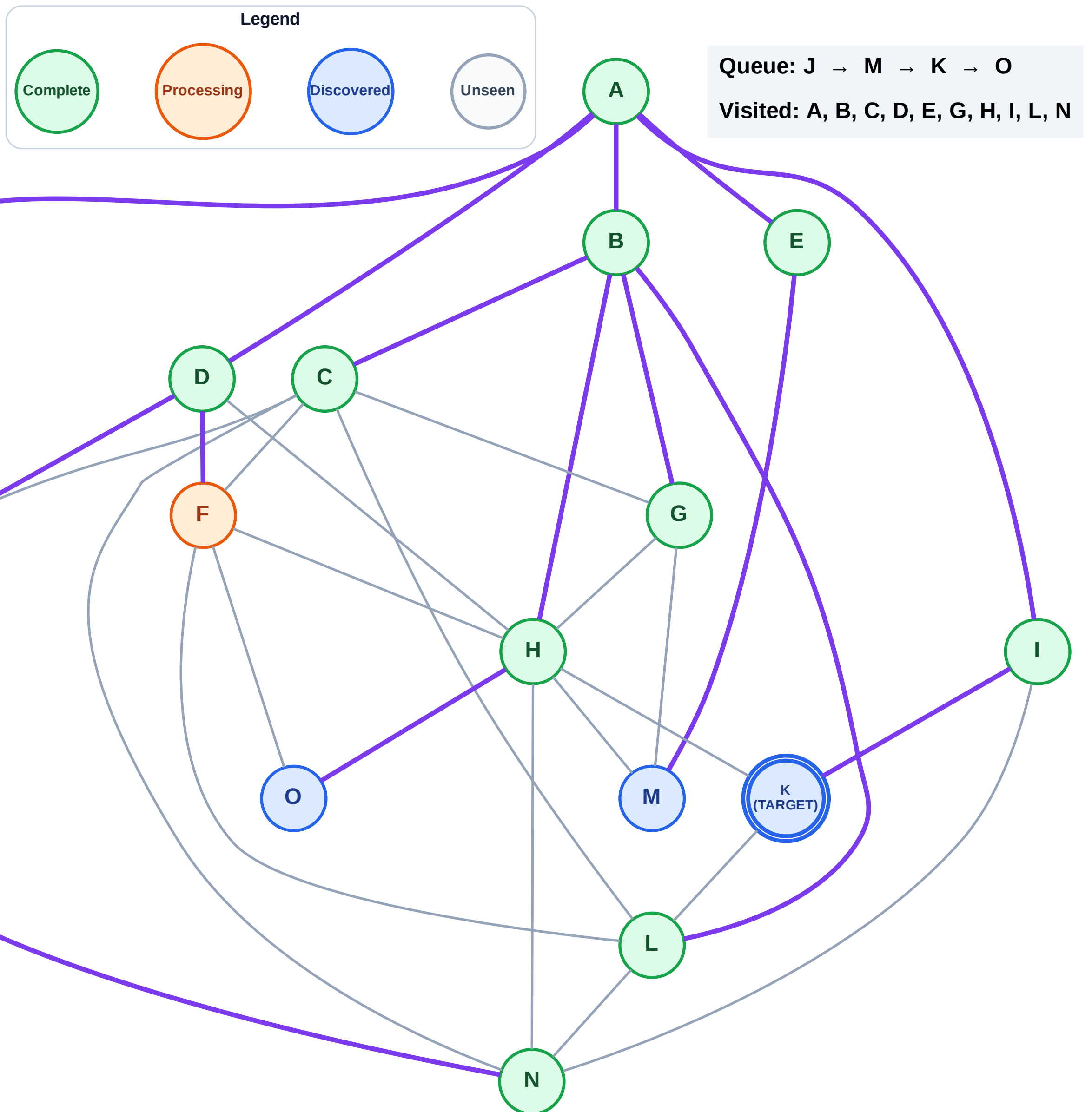


Queue: F → J → M → K → O

Visited: A, B, C, D, E, G, H, I, L, N



Step 22: Process node F



BFS Traversal

Step 23: No new neighbors from F

Legend



Complete



Processing



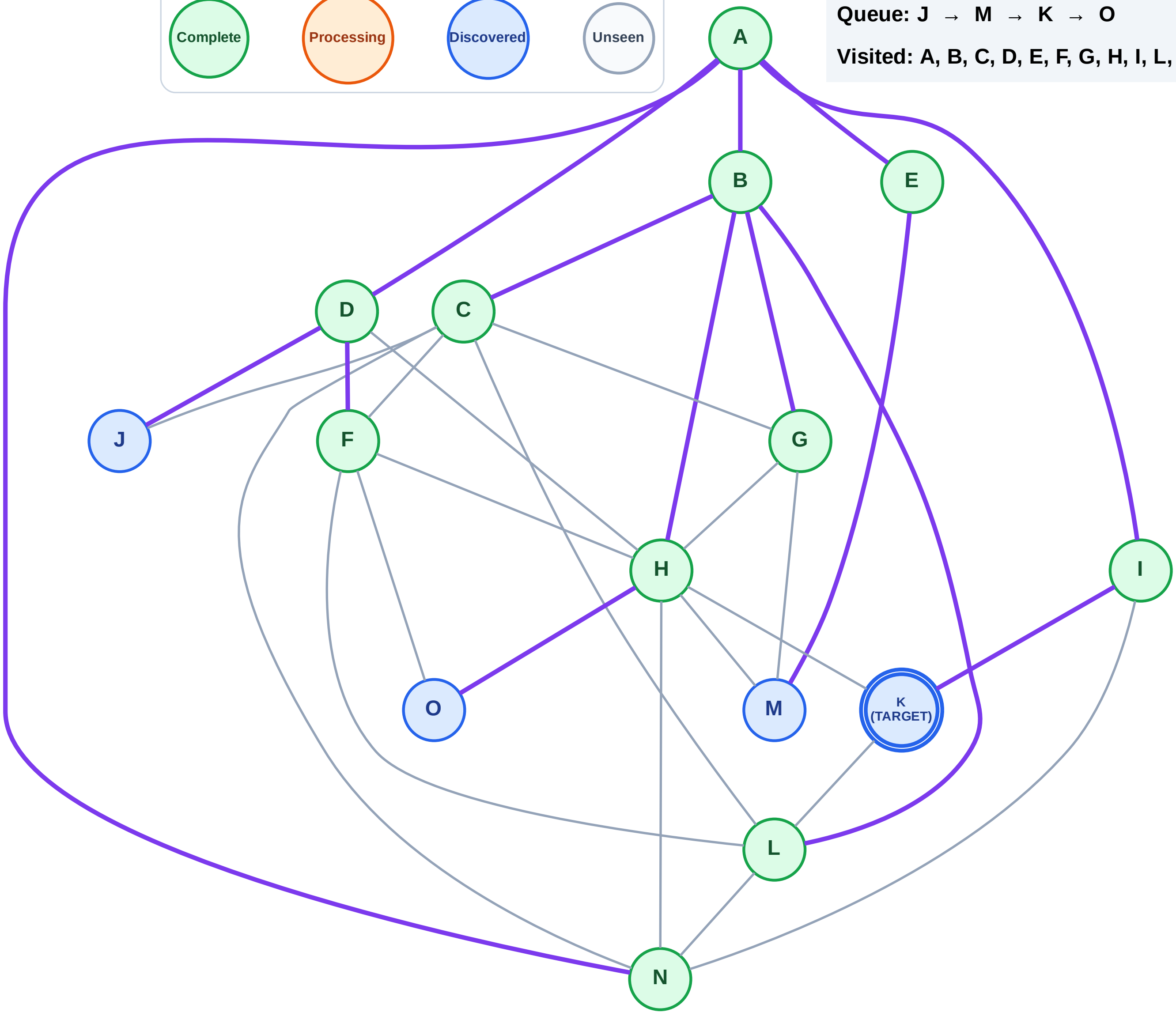
Discovered



Unseen

Queue: J → M → K → O

Visited: A, B, C, D, E, F, G, H, I, L, N



BFS Traversal

Step 24: Process node J

Legend

Complete

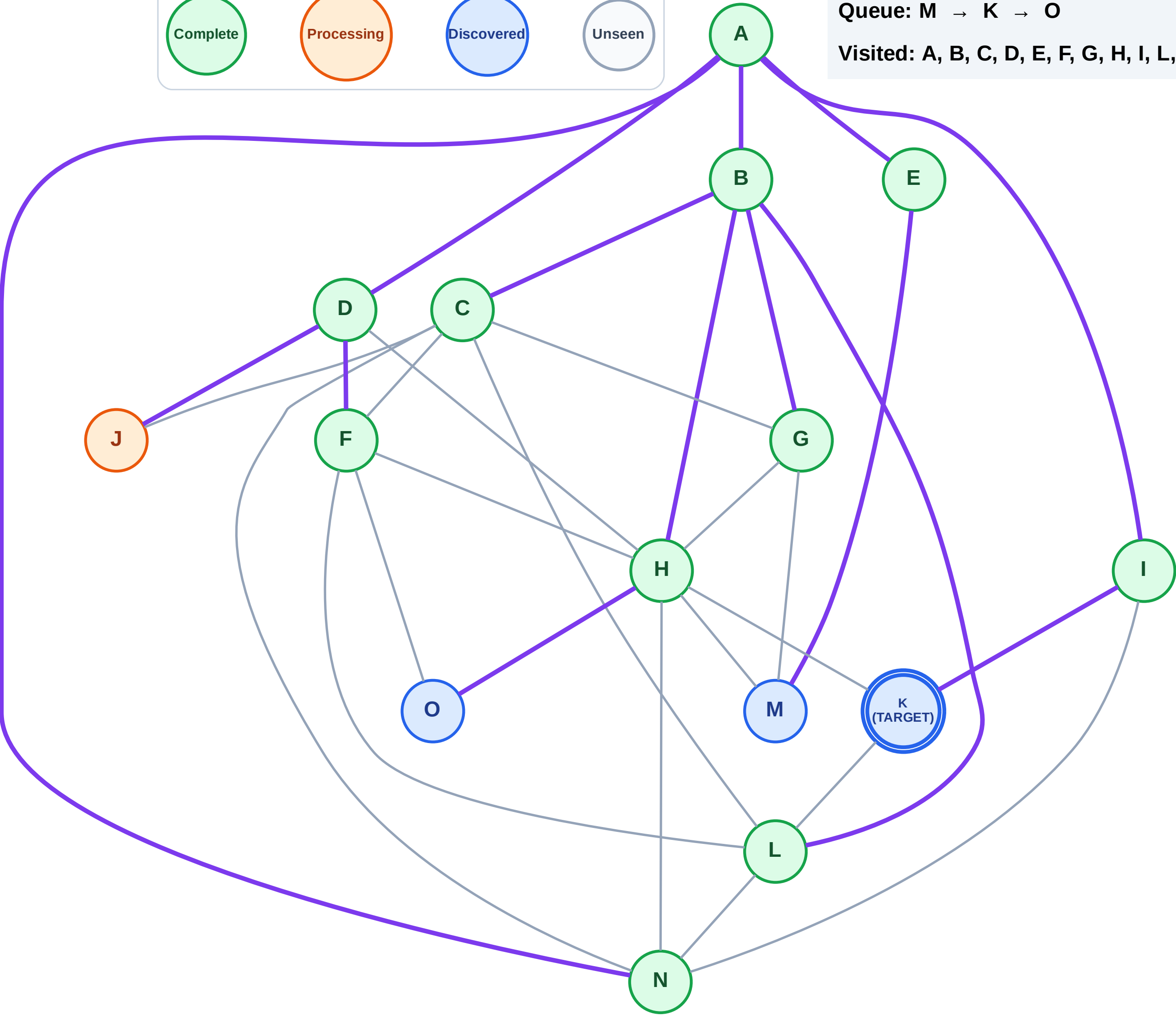
Processing

Discovered

Unseen

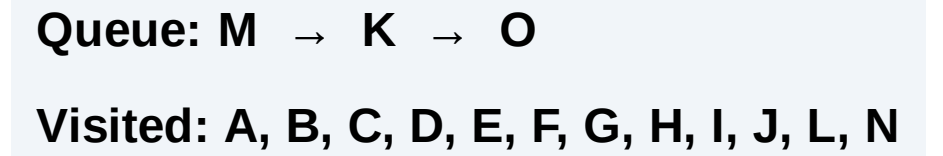
Queue: M → K → O

Visited: A, B, C, D, E, F, G, H, I, L, N



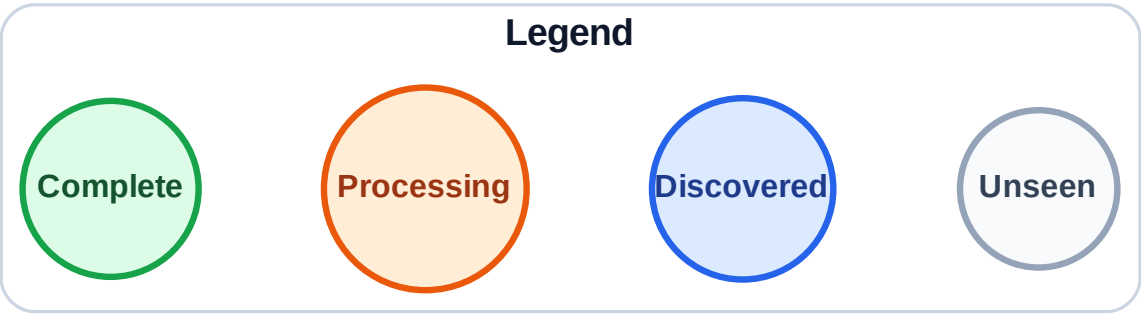
Step 25: No new neighbors from J

Step 25: No new neighbors from J



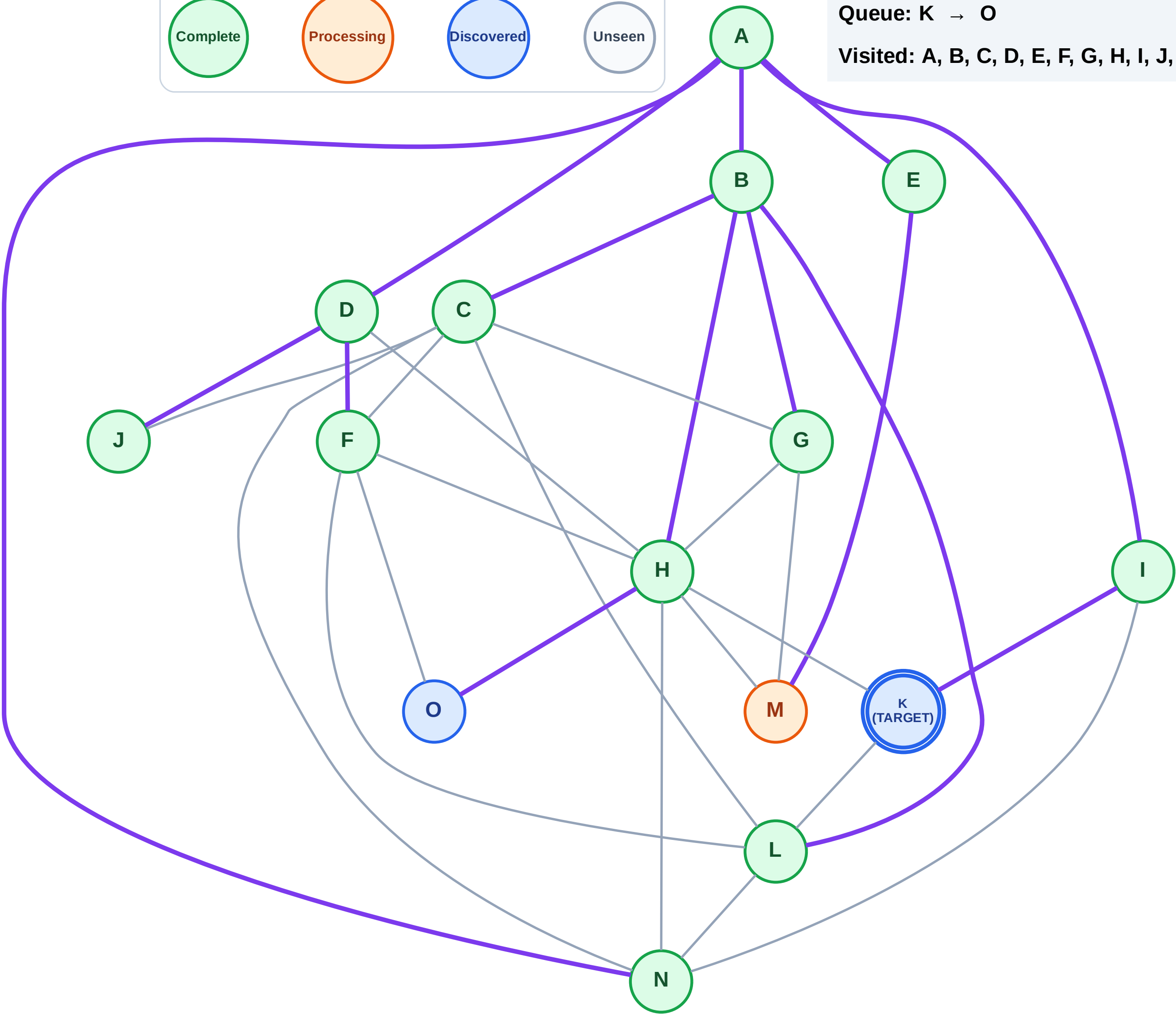
BFS Traversal

Step 26: Process node M



Queue: K → O

Visited: A, B, C, D, E, F, G, H, I, J, L, N



BFS Traversal

Step 27: No new neighbors from M

Legend

Complete

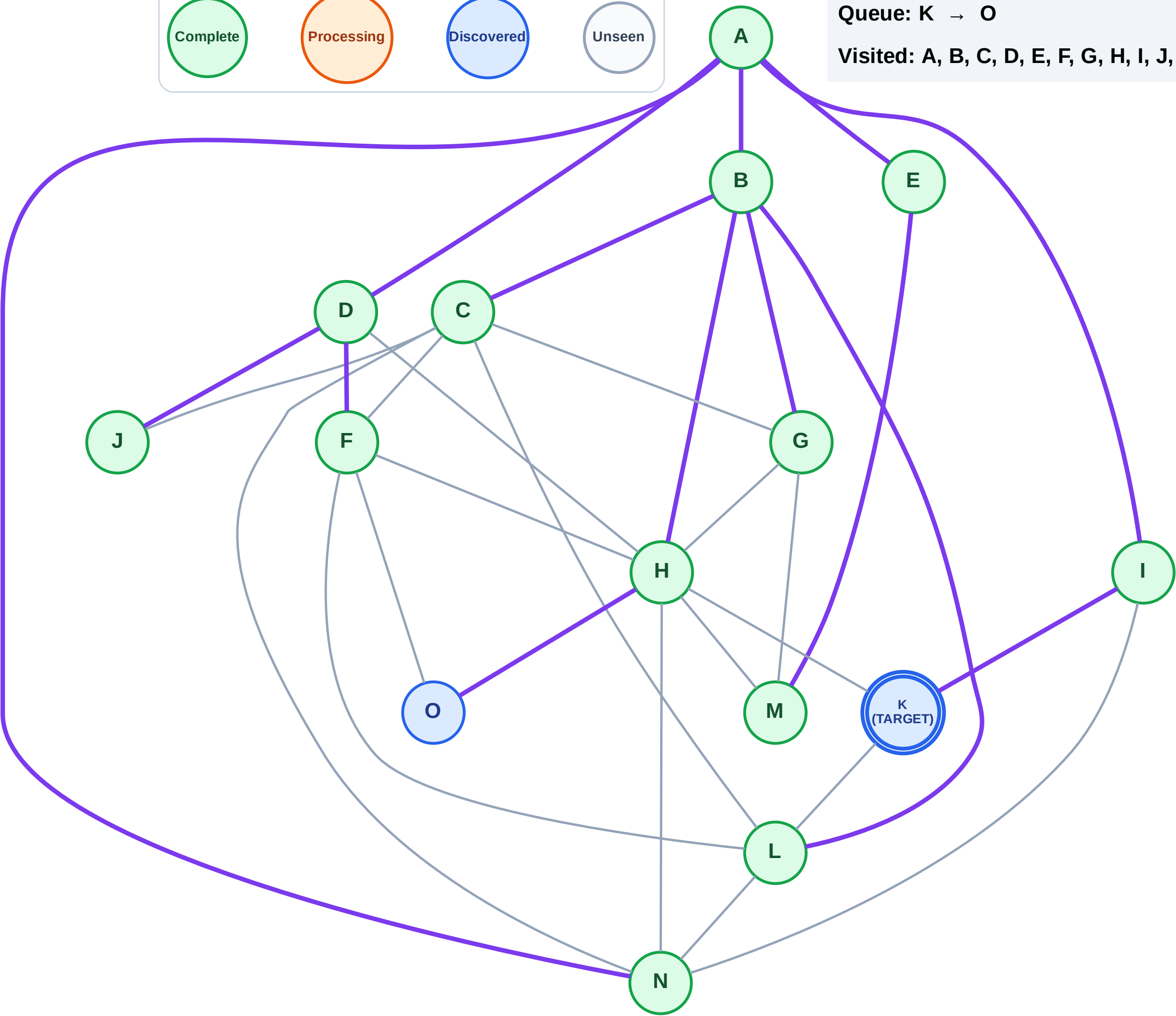
Processing

Discovered

Unseen

Queue: K → O

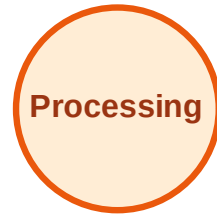
Visited: A, B, C, D, E, F, G, H, I, J, L, M, N



BFS Traversal

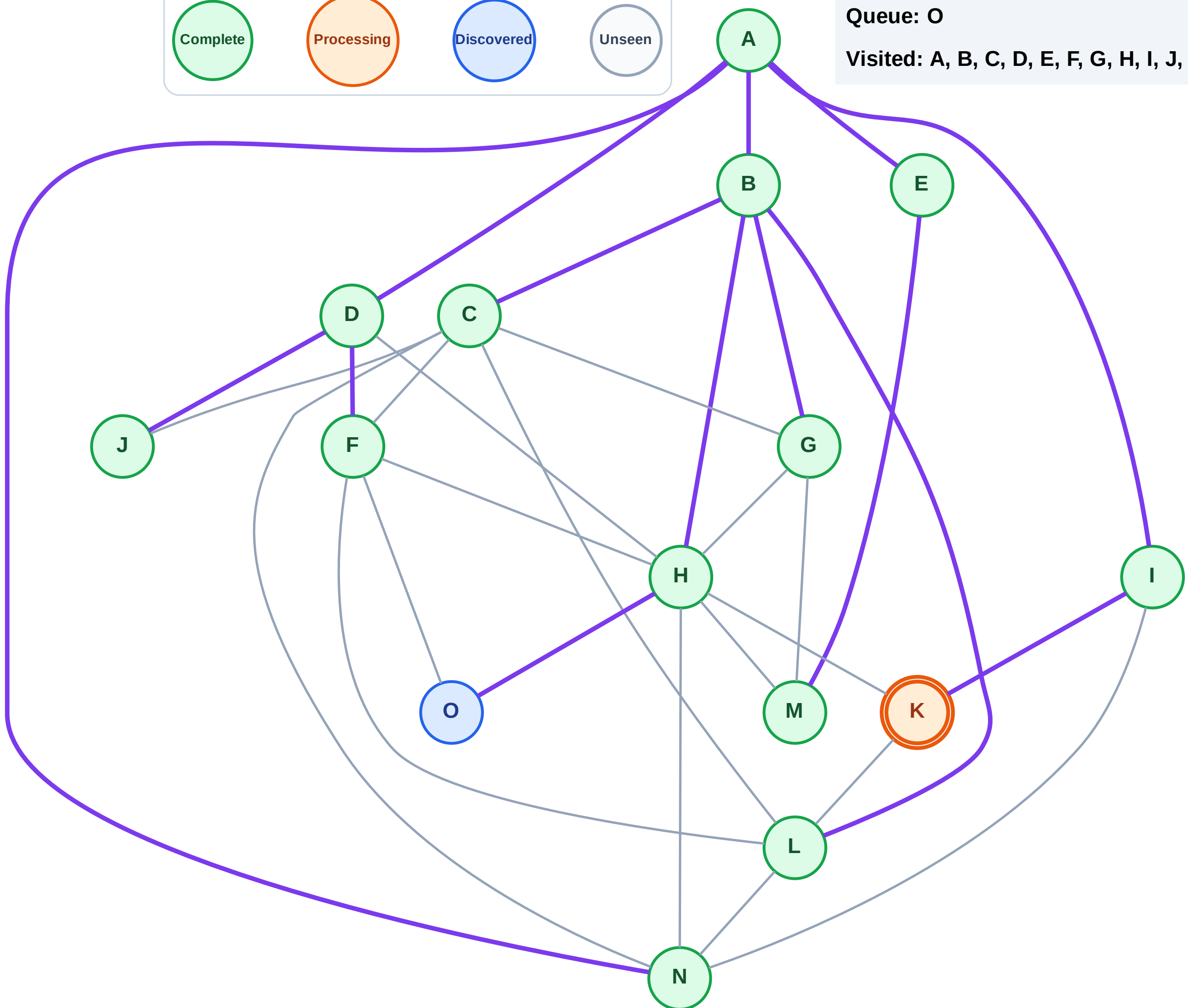
Step 28: Process node K

Legend



Queue: O

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N



BFS Traversal

Step 29: TARGET NODE K REACHED! Path (2 hops): A → I → K

Legend

Complete

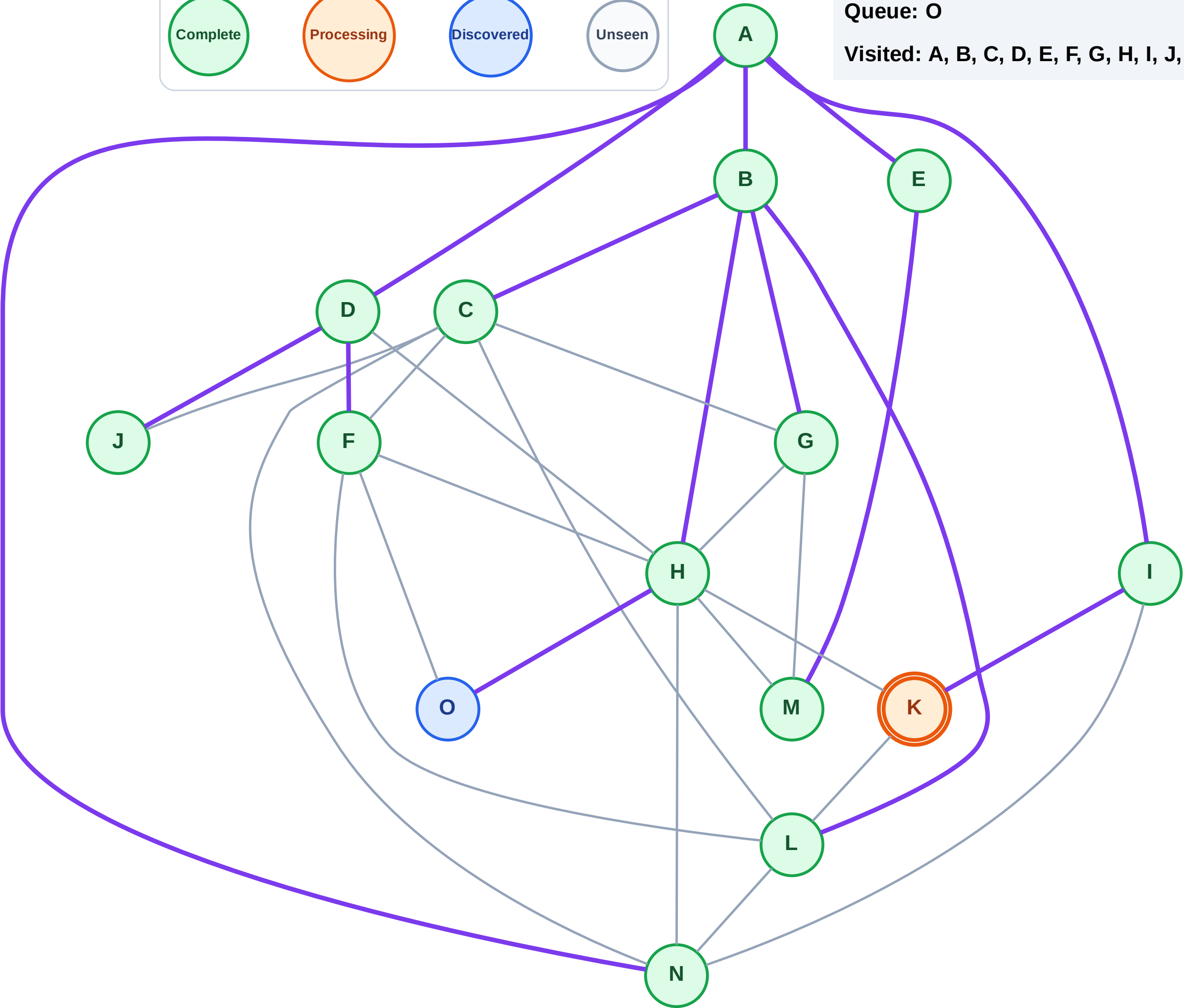
Processing

Discovered

Unseen

Queue: O

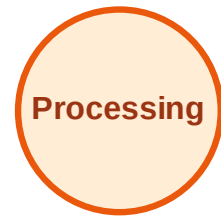
Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N



DFS Traversal

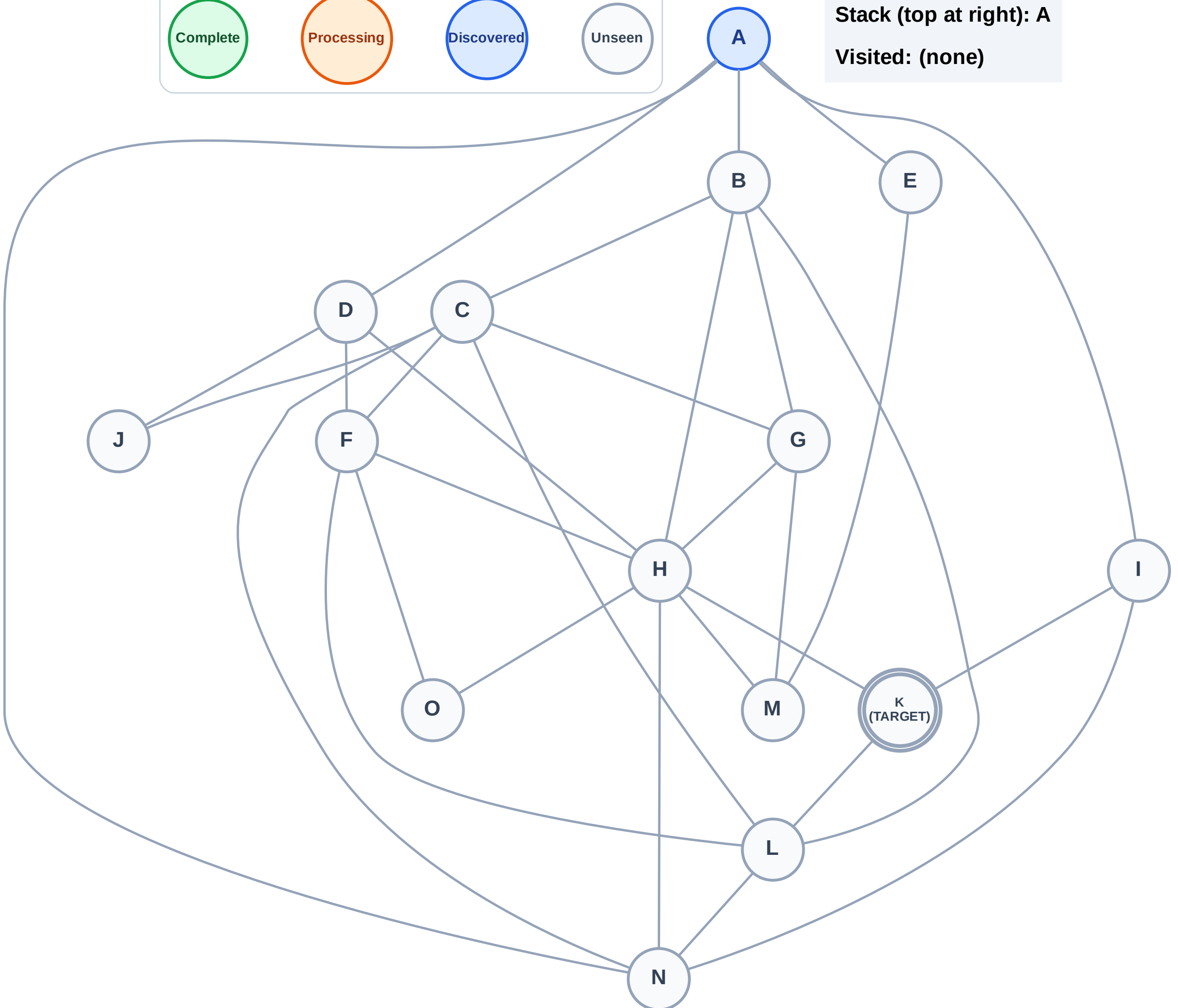
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

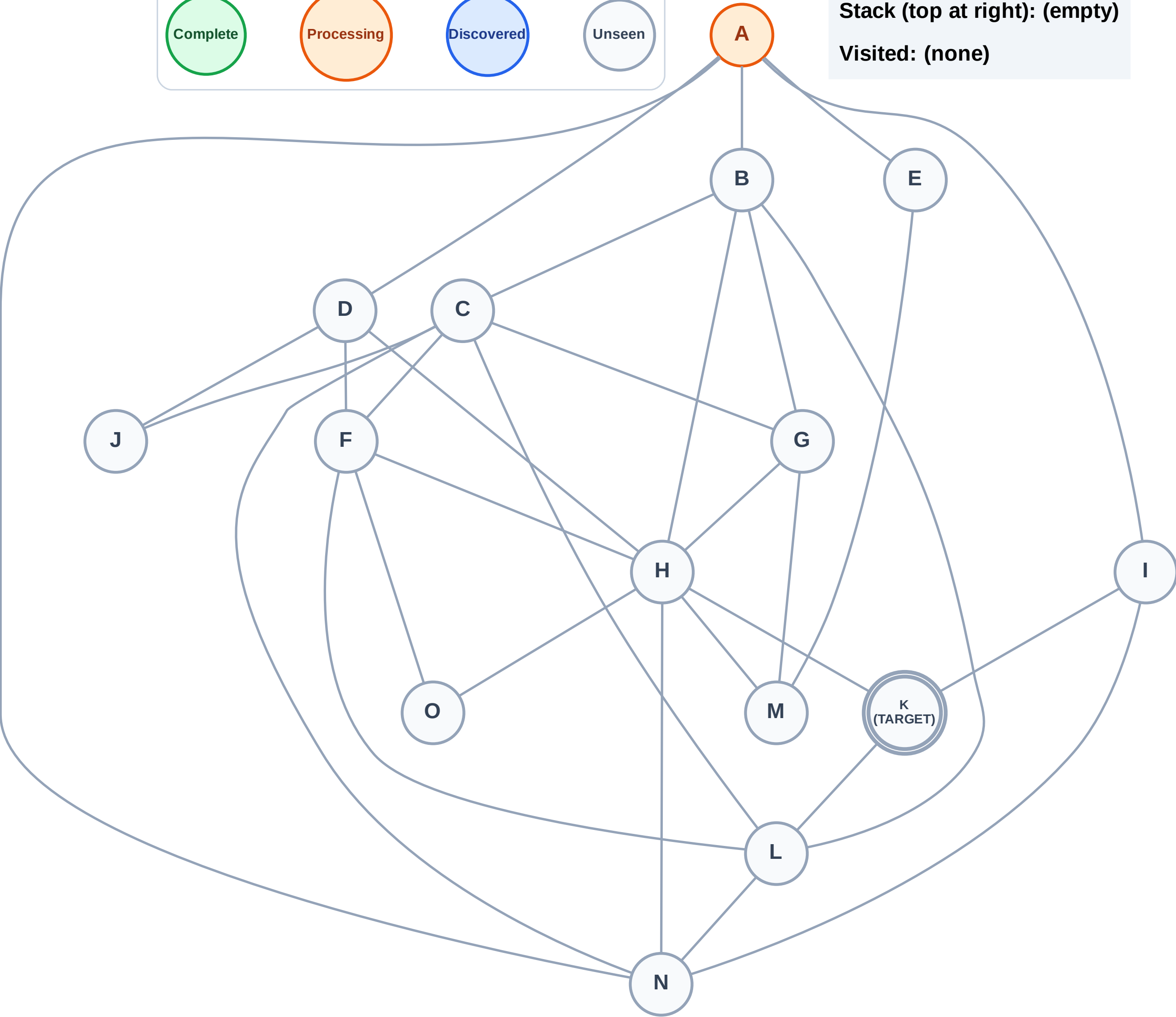
Complete

Processing

Discovered

Unseen

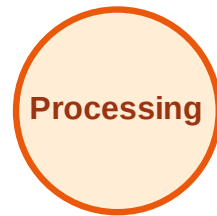
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

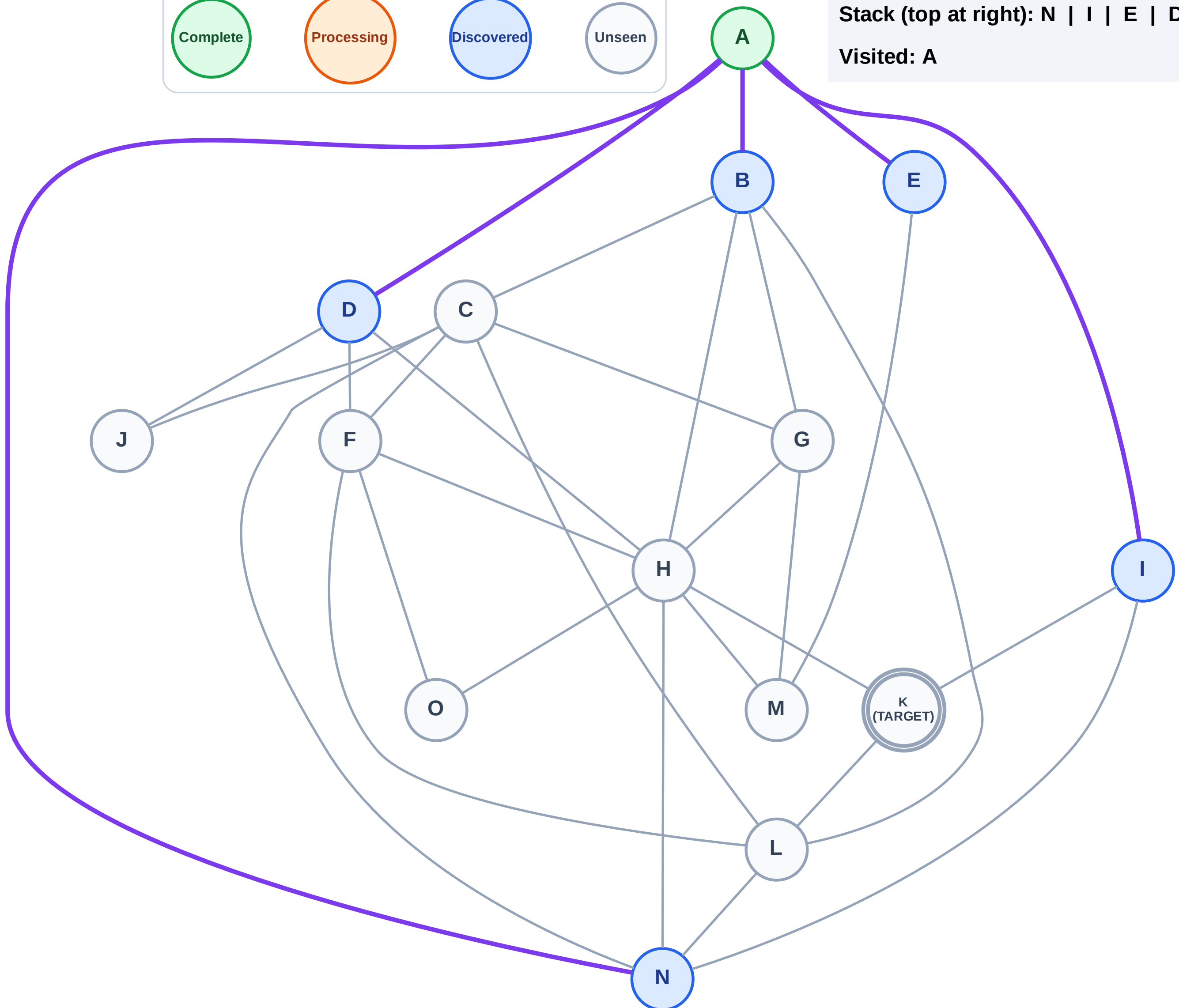
Step 3: Discover N, I, E, D, B from A

Legend



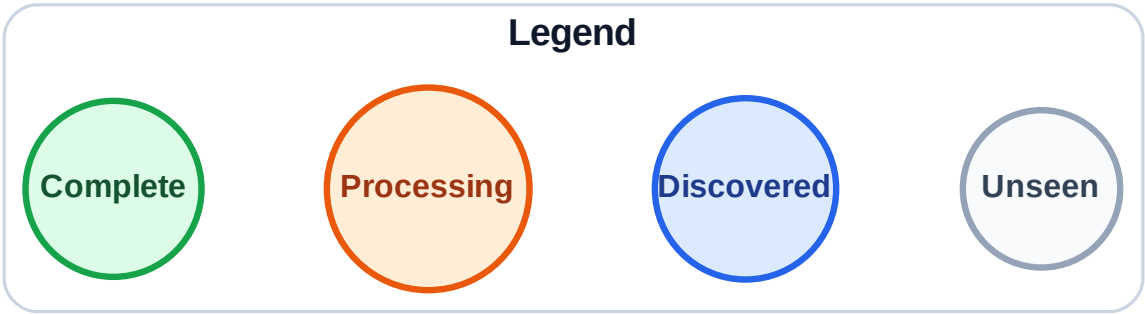
Stack (top at right): N | I | E | D | B

Visited: A



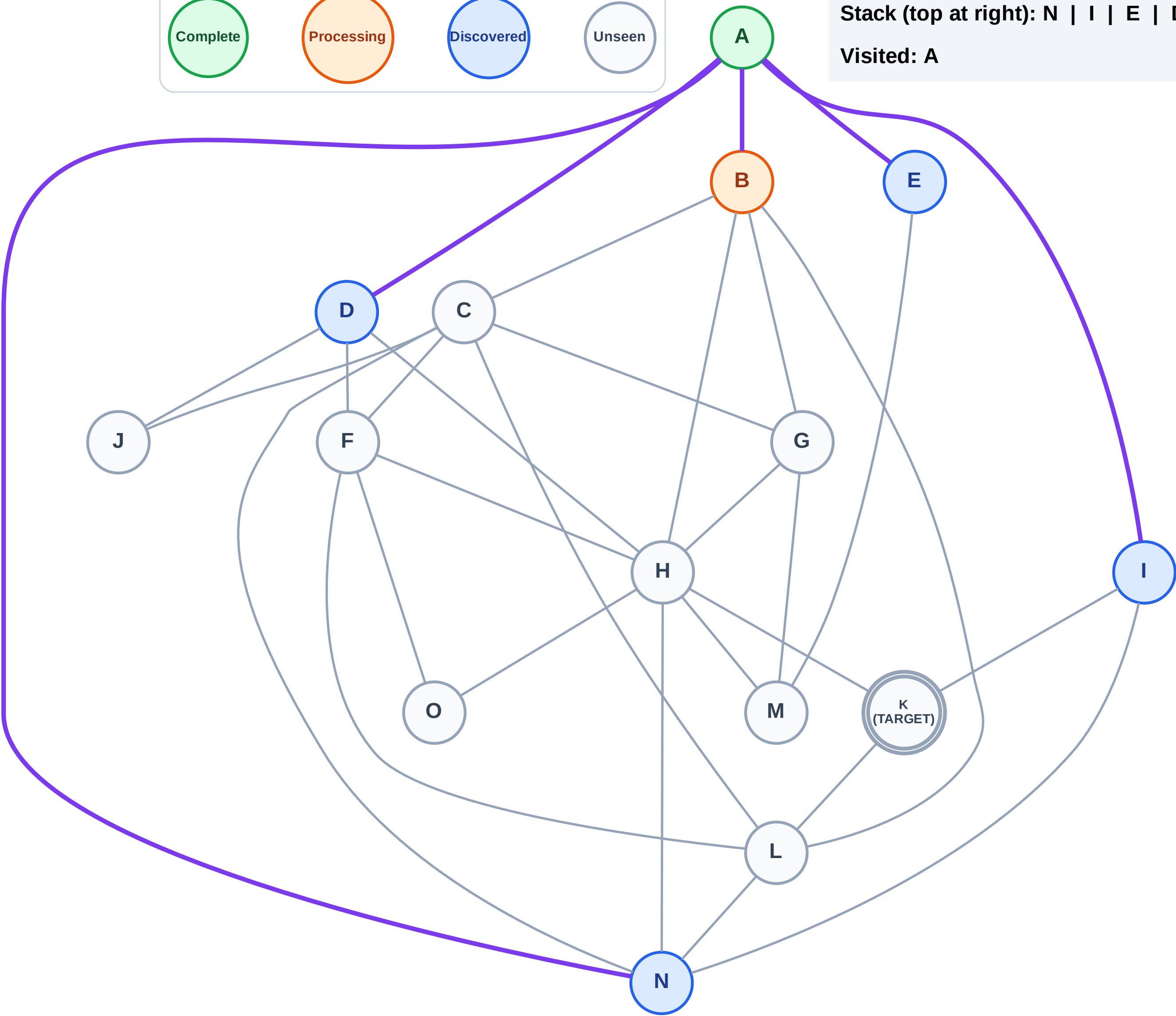
DFS Traversal

Step 4: Process node B



Stack (top at right): N | I | E | D

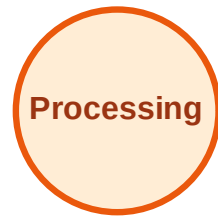
Visited: A



DFS Traversal

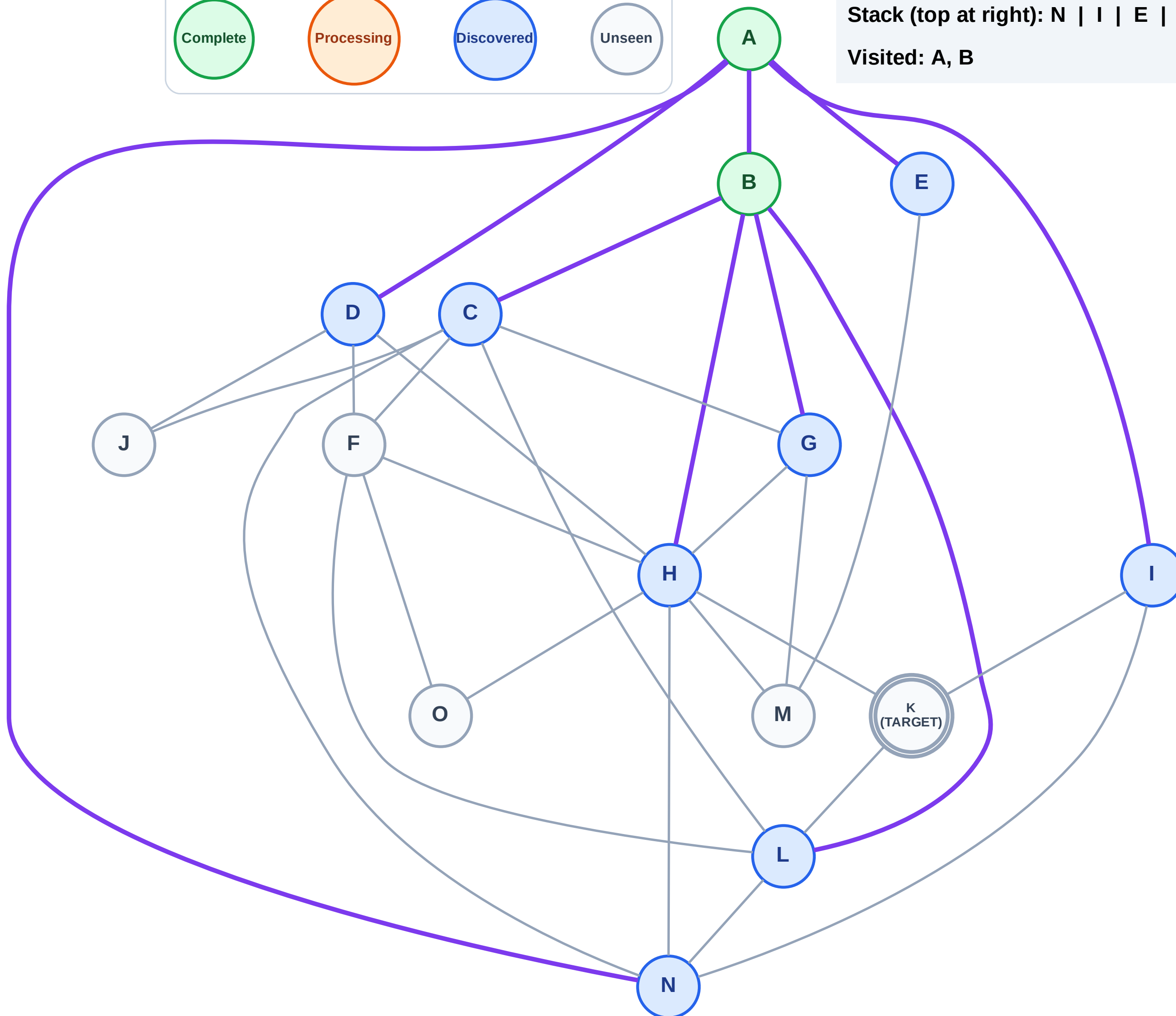
Step 5: Discover L, H, G, C from B

Legend



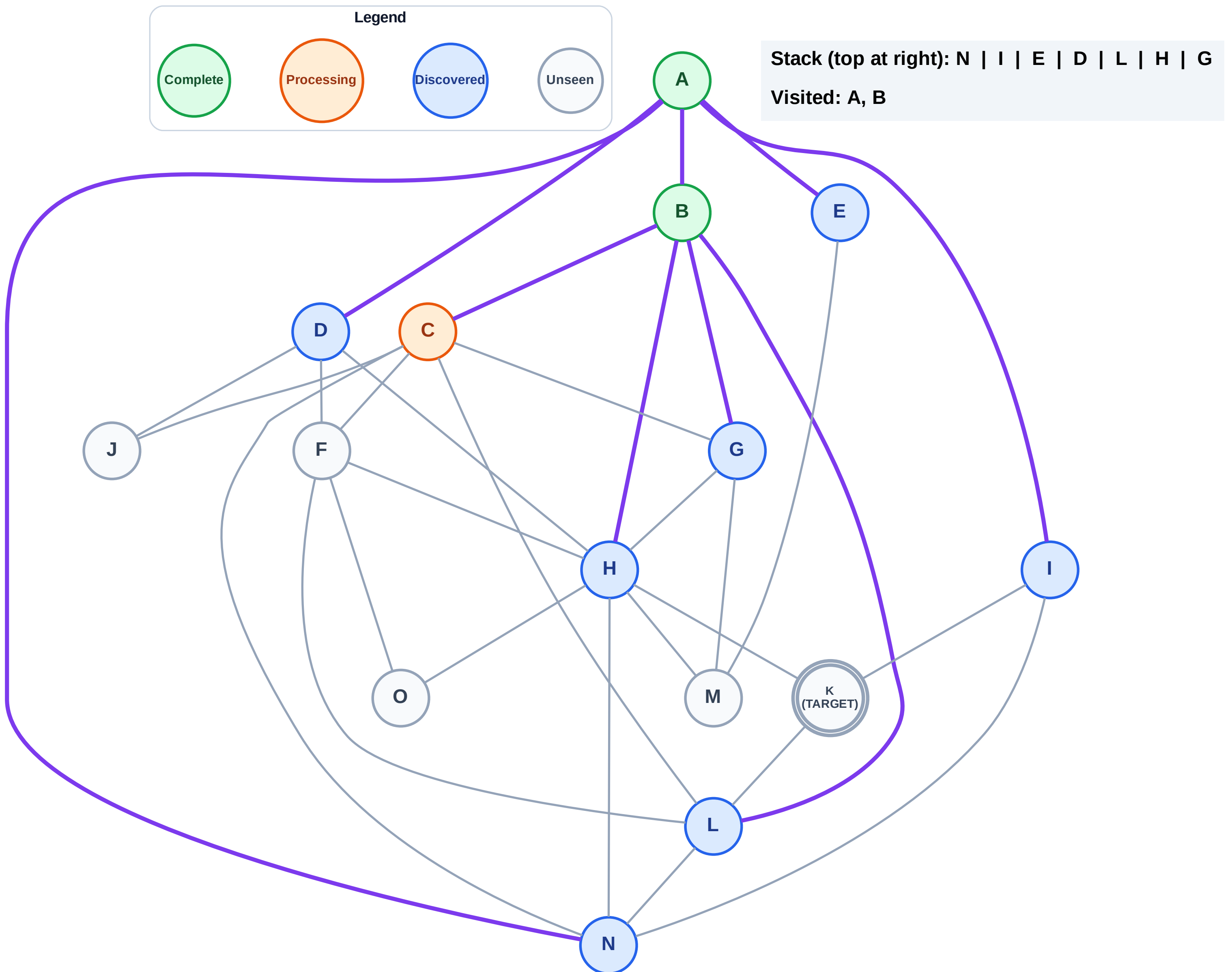
Stack (top at right): N | I | E | D | L | H | G | C

Visited: A, B



Step 6: Process node C

Step 6: Process node C



DFS Traversal

Step 7: Discover J, F from C

Legend

Complete

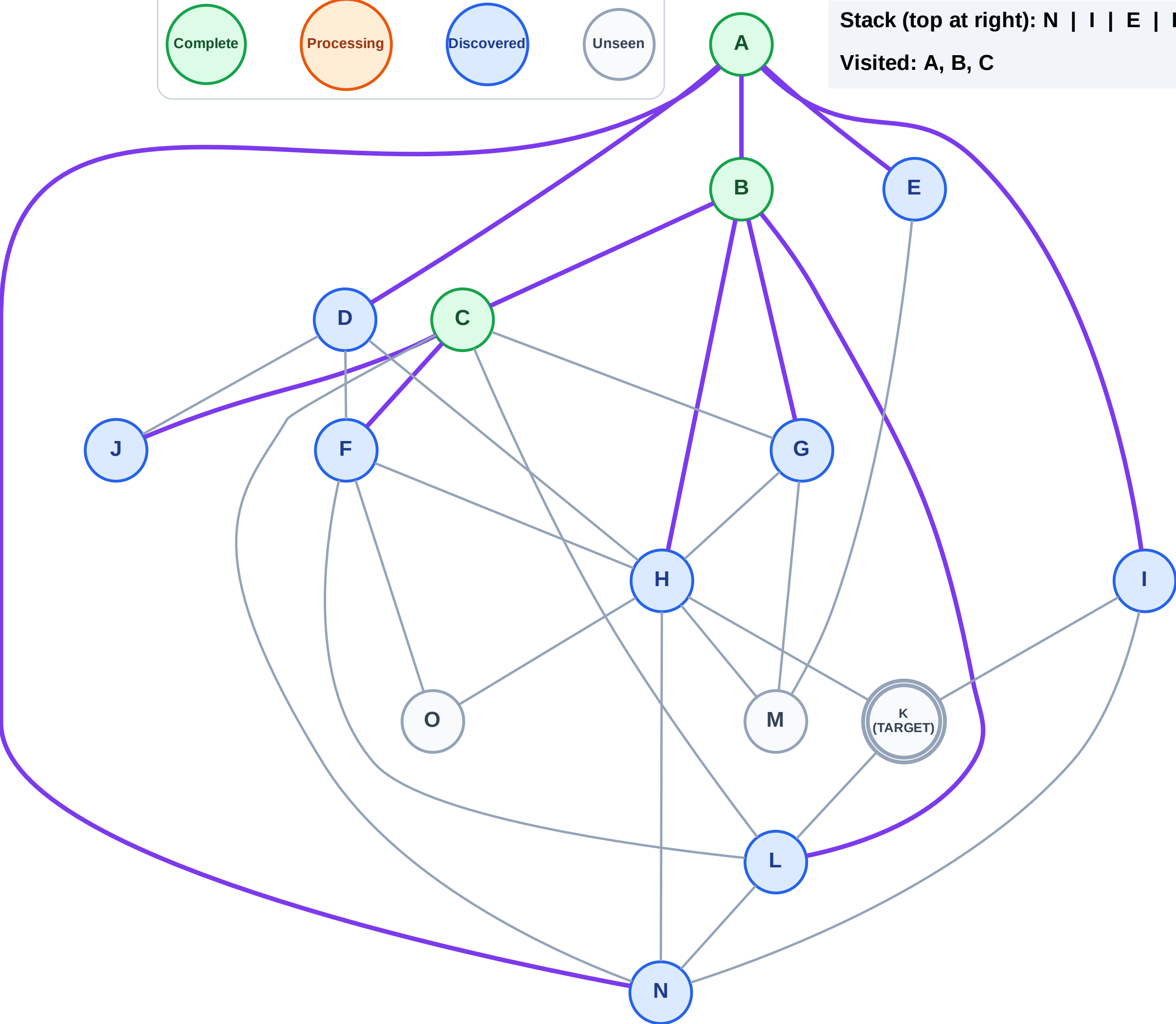
Processing

Discovered

Unseen

Stack (top at right): N | I | E | D | L | H | G | J | F

Visited: A, B, C



DFS Traversal

Step 8: Process node F

Legend

Complete

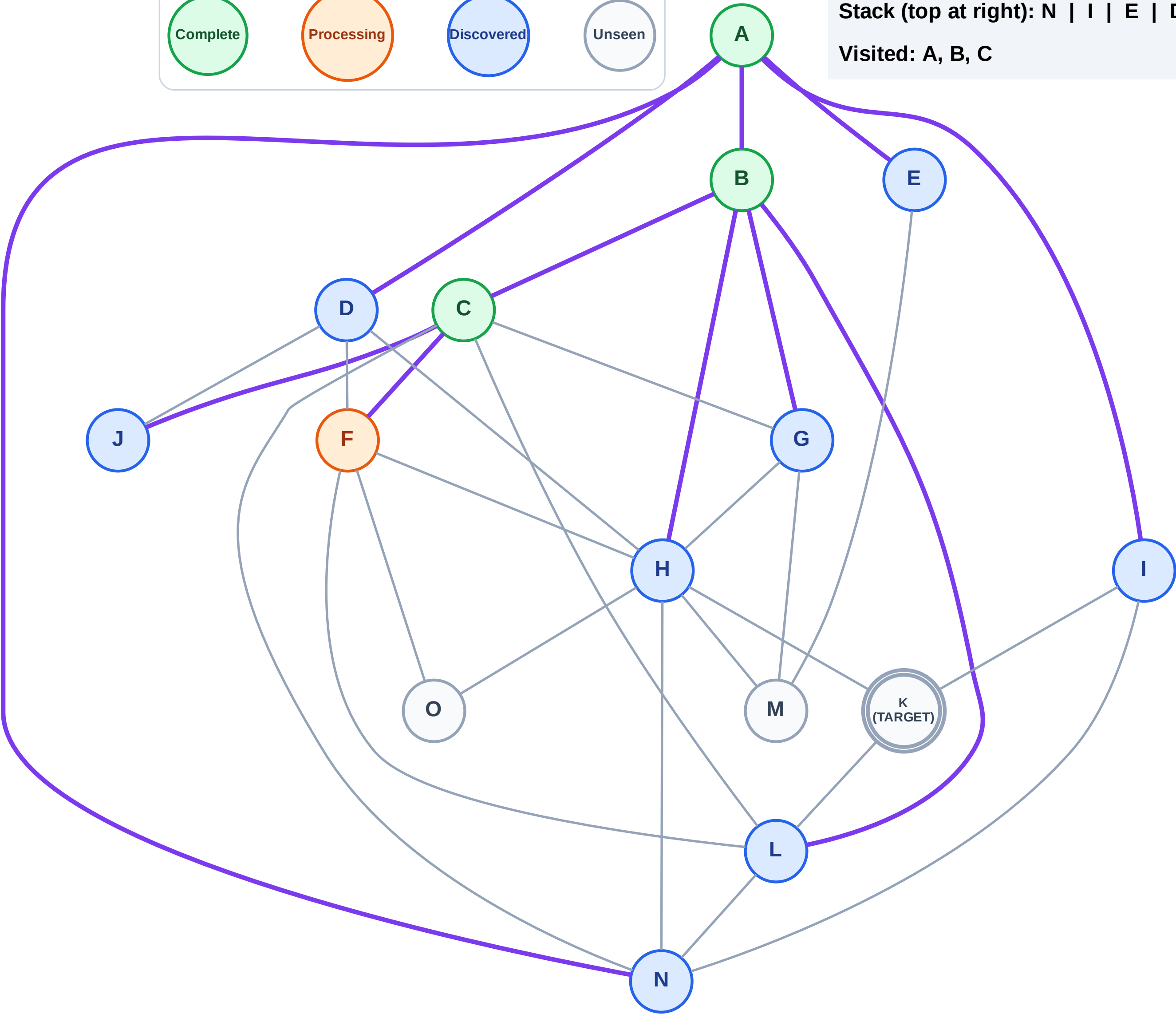
Processing

Discovered

Unseen

Stack (top at right): N | I | E | D | L | H | G | J

Visited: A, B, C



DFS Traversal

Step 9: Discover O from F

Legend

Complete

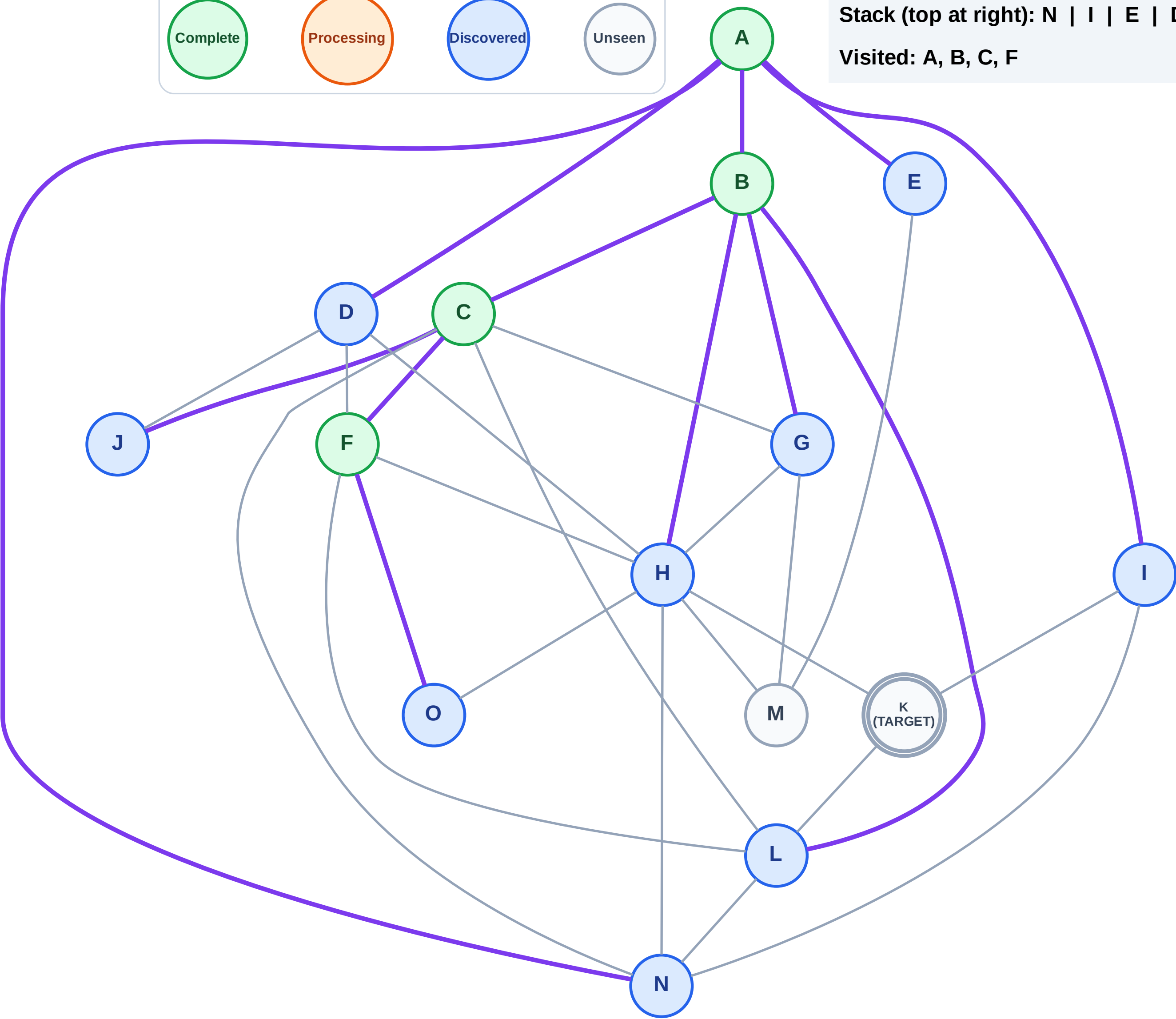
Processing

Discovered

Unseen

Stack (top at right): N | I | E | D | L | H | G | J | O

Visited: A, B, C, F



DFS Traversal

Step 10: Process node O

Legend

Complete

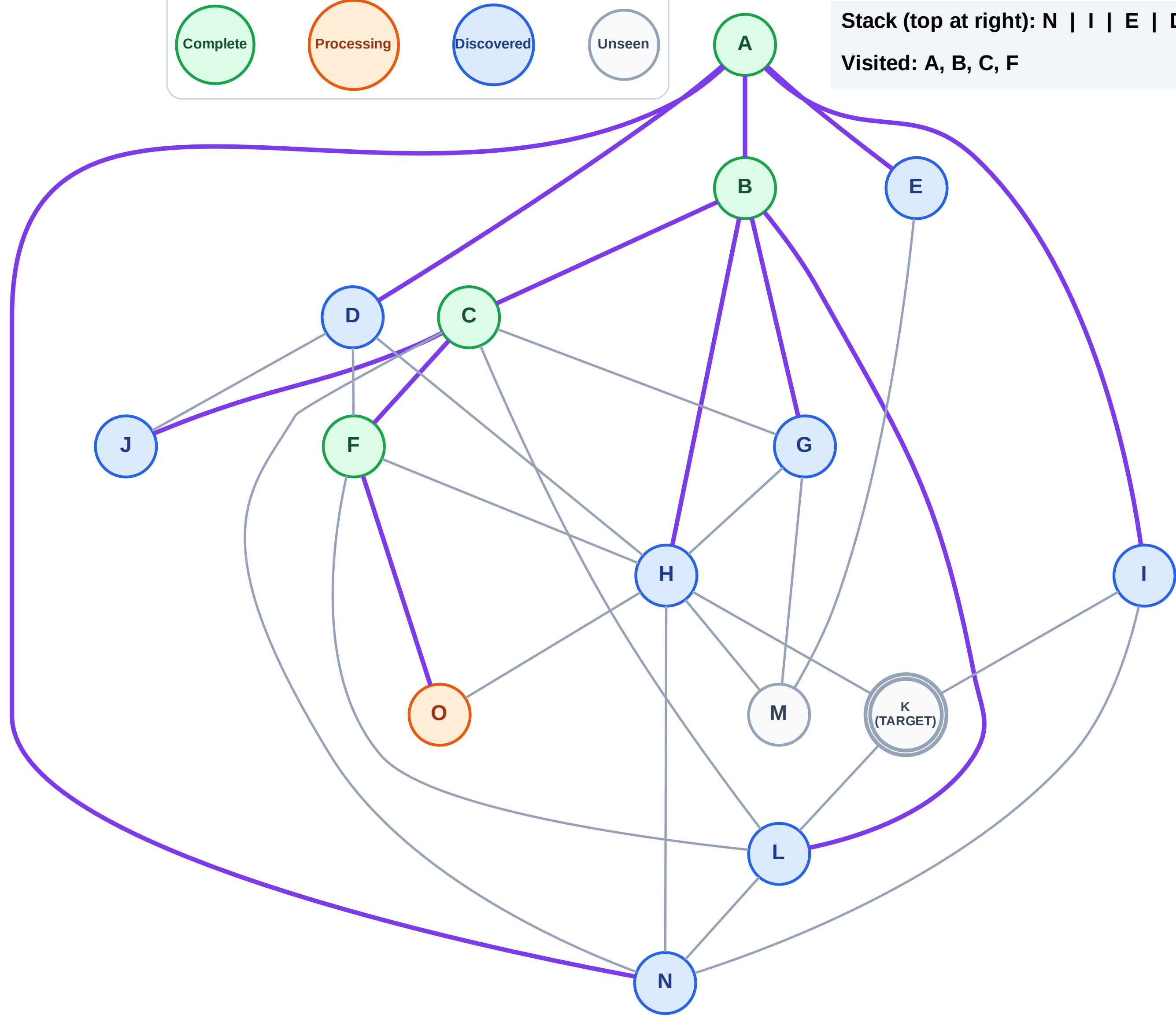
Processing

Discovered

Unseen

Stack (top at right): N | I | E | D | L | H | G | J

Visited: A, B, C, F



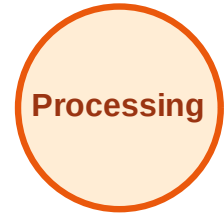
DFS Traversal

Step 11: No new neighbors from O

Legend



Complete



Processing



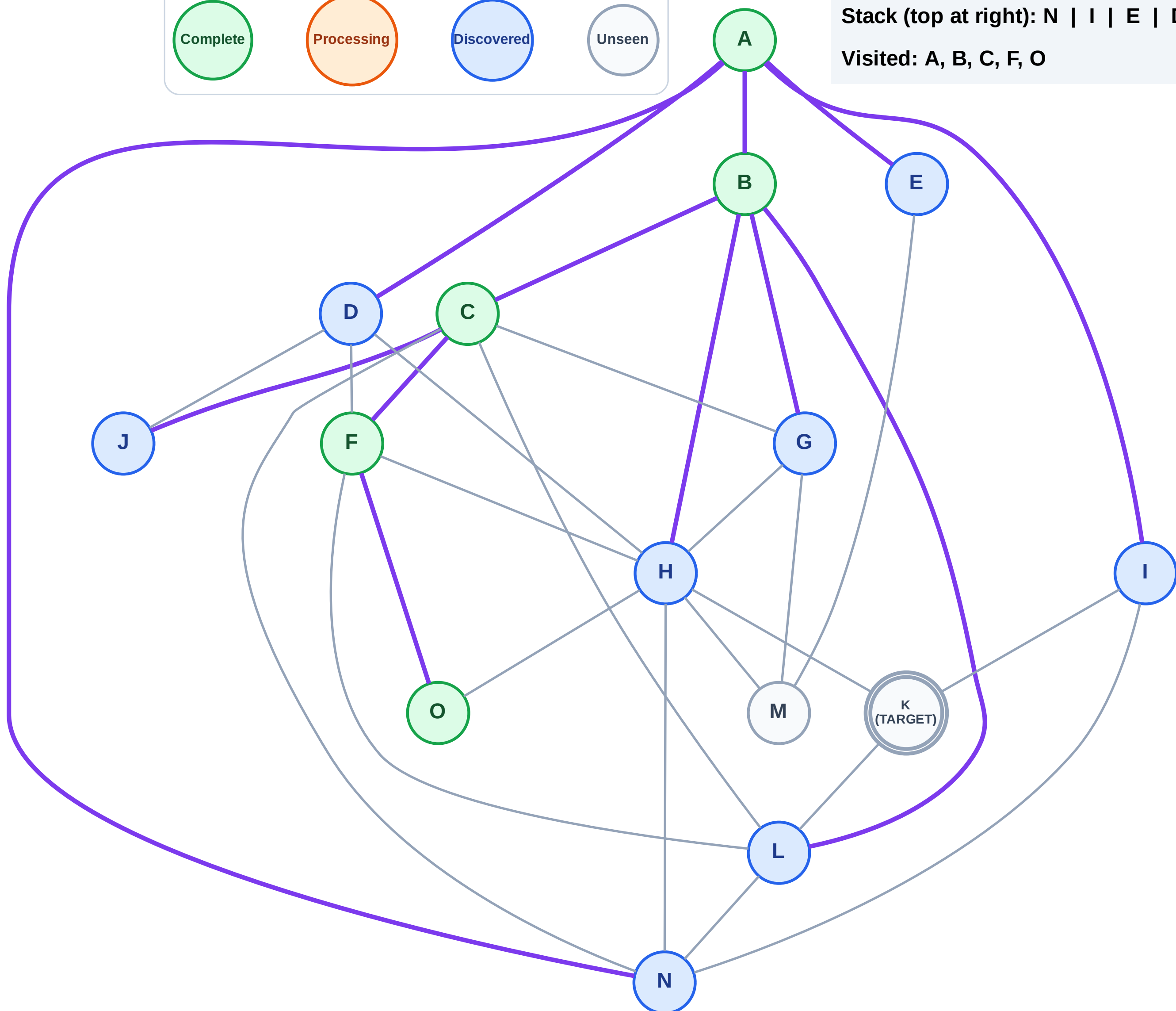
Discovered



Unseen

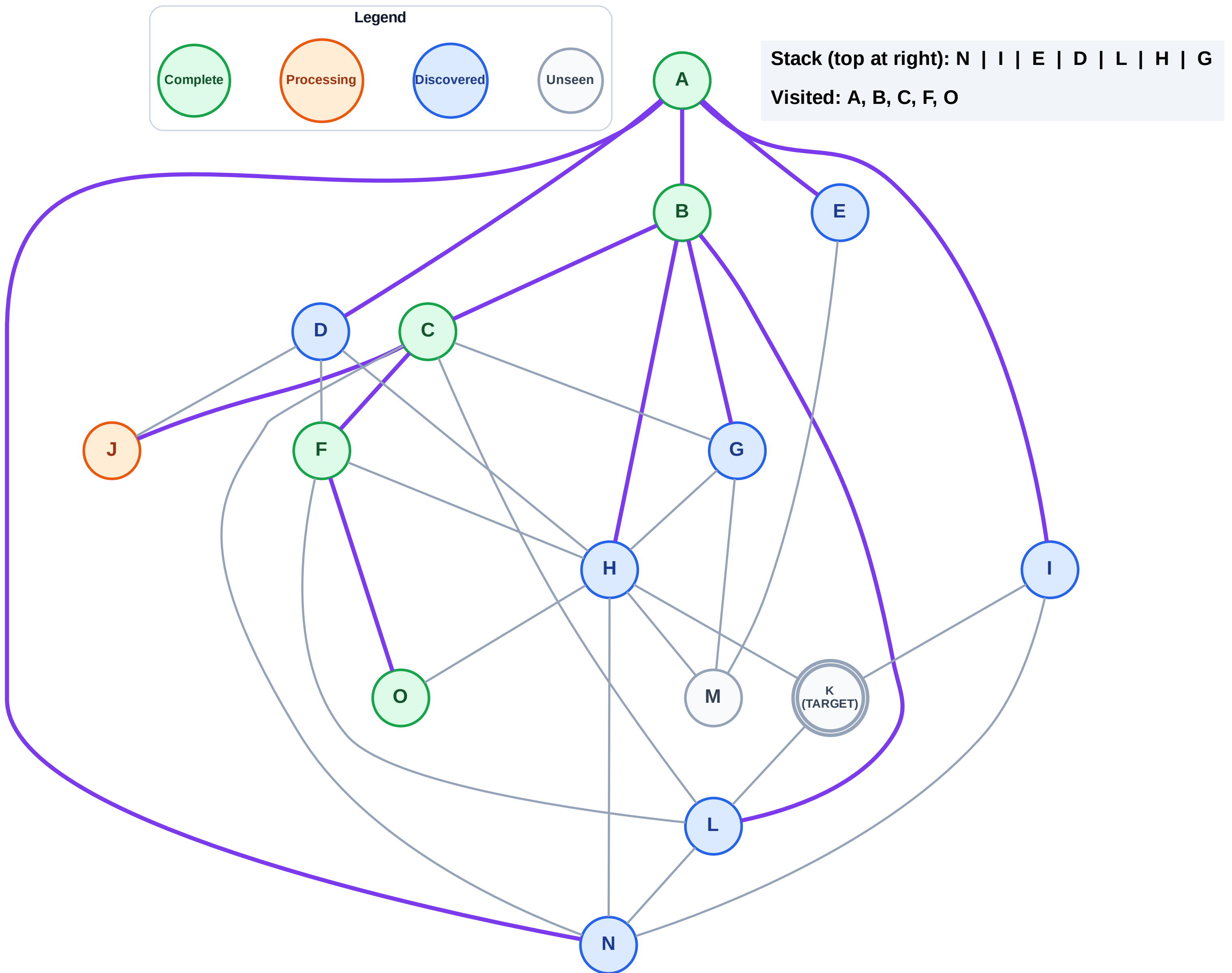
Stack (top at right): N | I | E | D | L | H | G | J

Visited: A, B, C, F, O



Step 12: Process node J

Step 12: Process node J

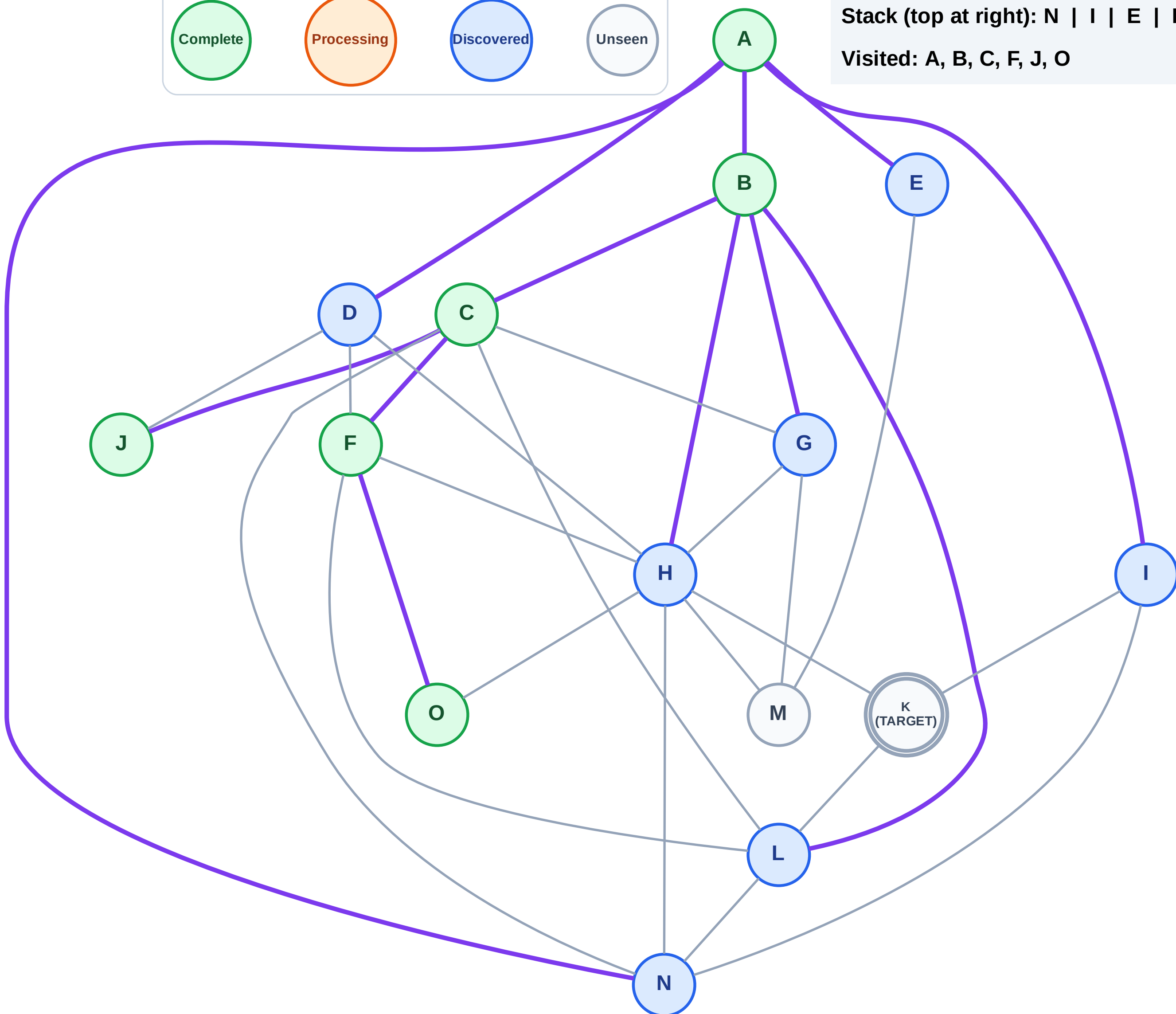


Step 13: No new neighbors from J

Legend

Unseen

Visited: A, B, C, F, J, O



DFS Traversal

Step 14: Process node G

Legend

Complete

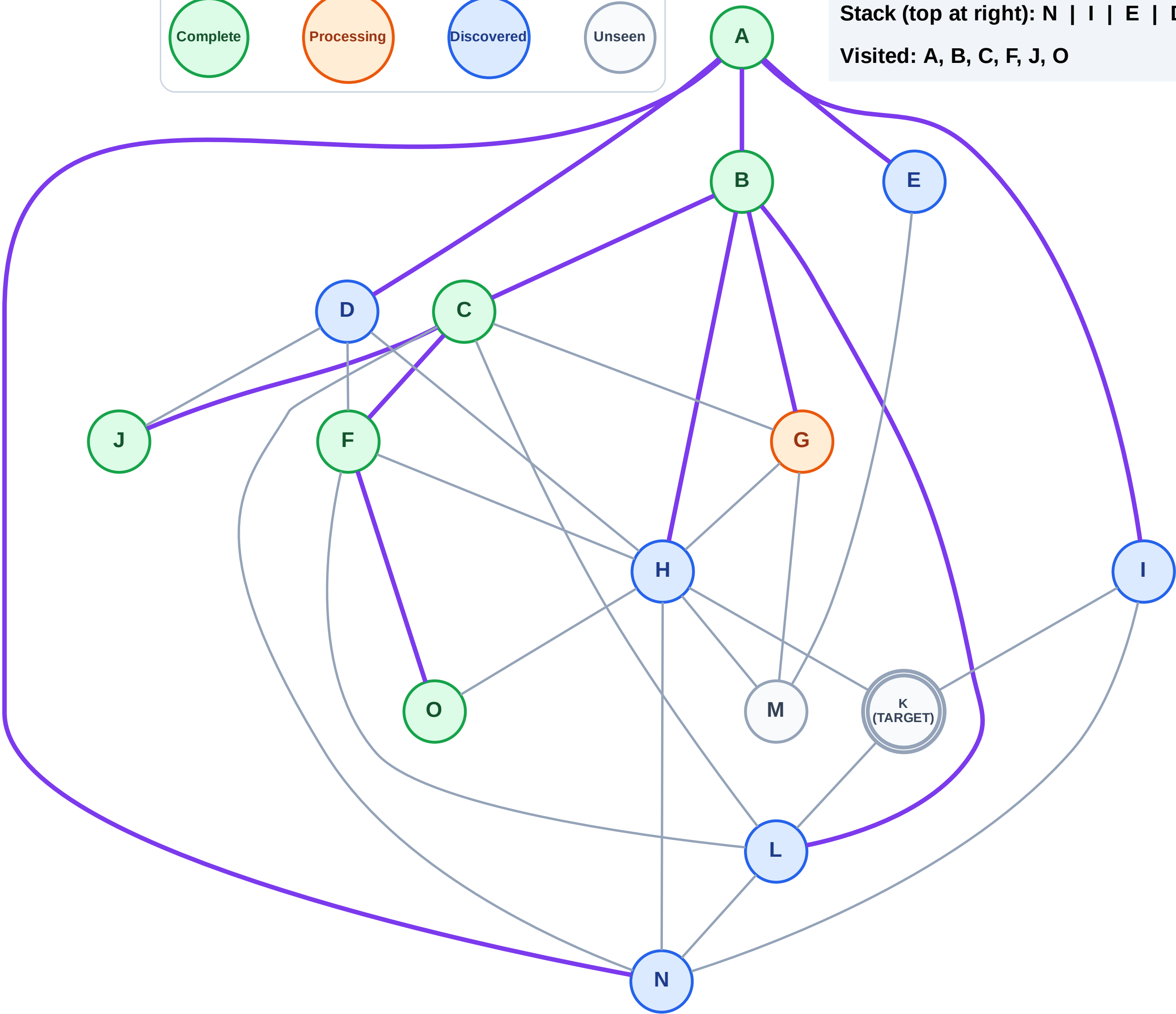
Processing

Discovered

Unseen

Stack (top at right): N | I | E | D | L | H

Visited: A, B, C, F, J, O



Step 15: Discover M from G

Step 15: Discover M from G



DFS Traversal

Step 16: Process node M

Legend

Complete

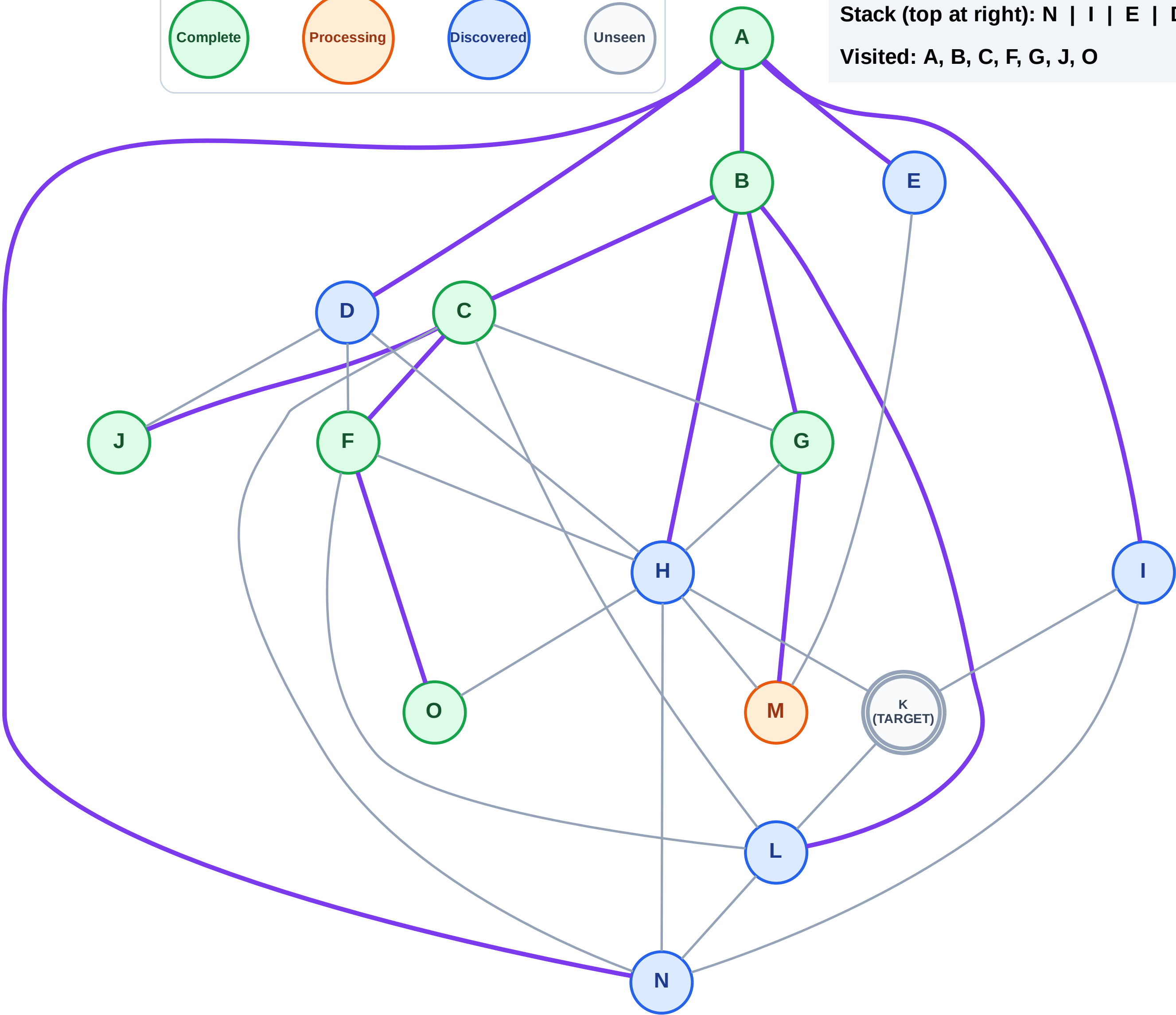
Processing

Discovered

Unseen

Stack (top at right): N | I | E | D | L | H

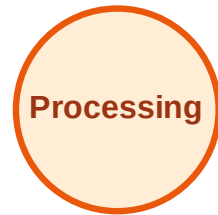
Visited: A, B, C, F, G, J, O



DFS Traversal

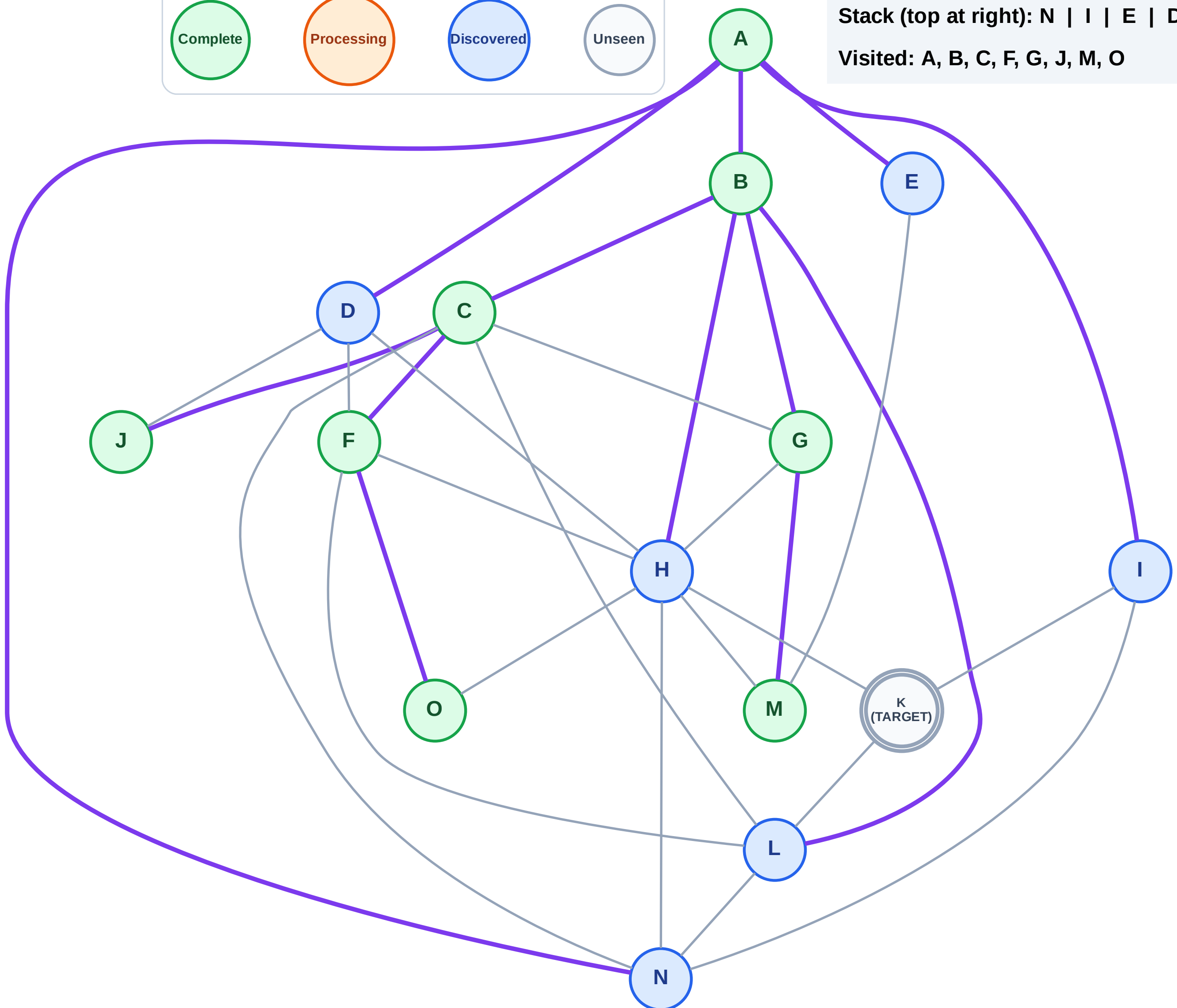
Step 17: No new neighbors from M

Legend



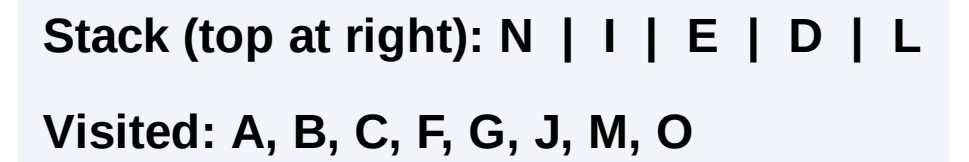
Stack (top at right): N | I | E | D | L | H

Visited: A, B, C, F, G, J, M, O



Step 18: Process node H

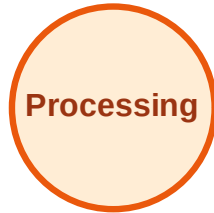
Step 18: Process node H



DFS Traversal

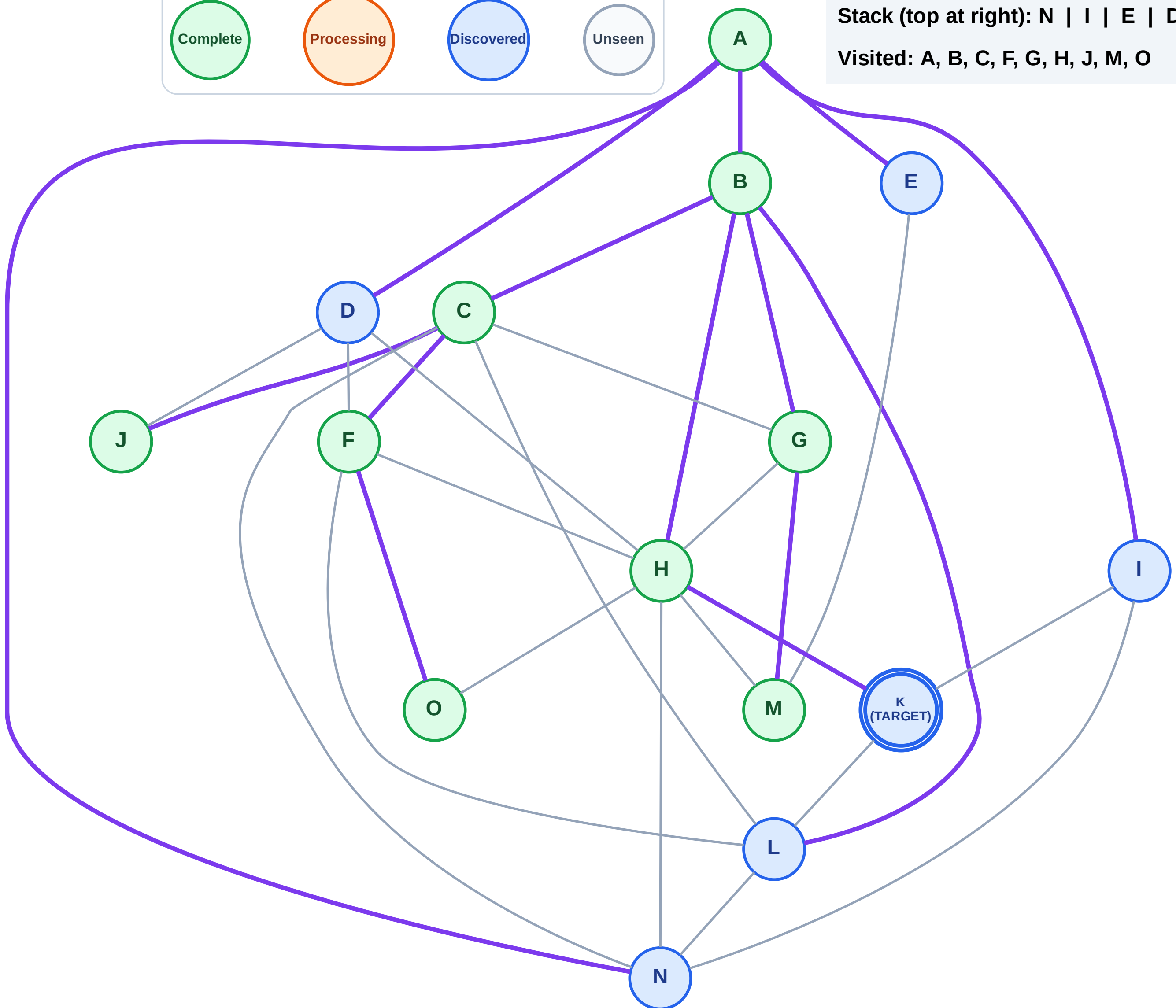
Step 19: Discover K from H

Legend



Stack (top at right): N | I | E | D | L | K

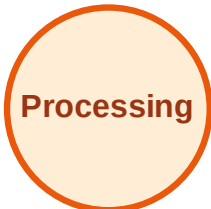
Visited: A, B, C, F, G, H, J, M, O



DFS Traversal

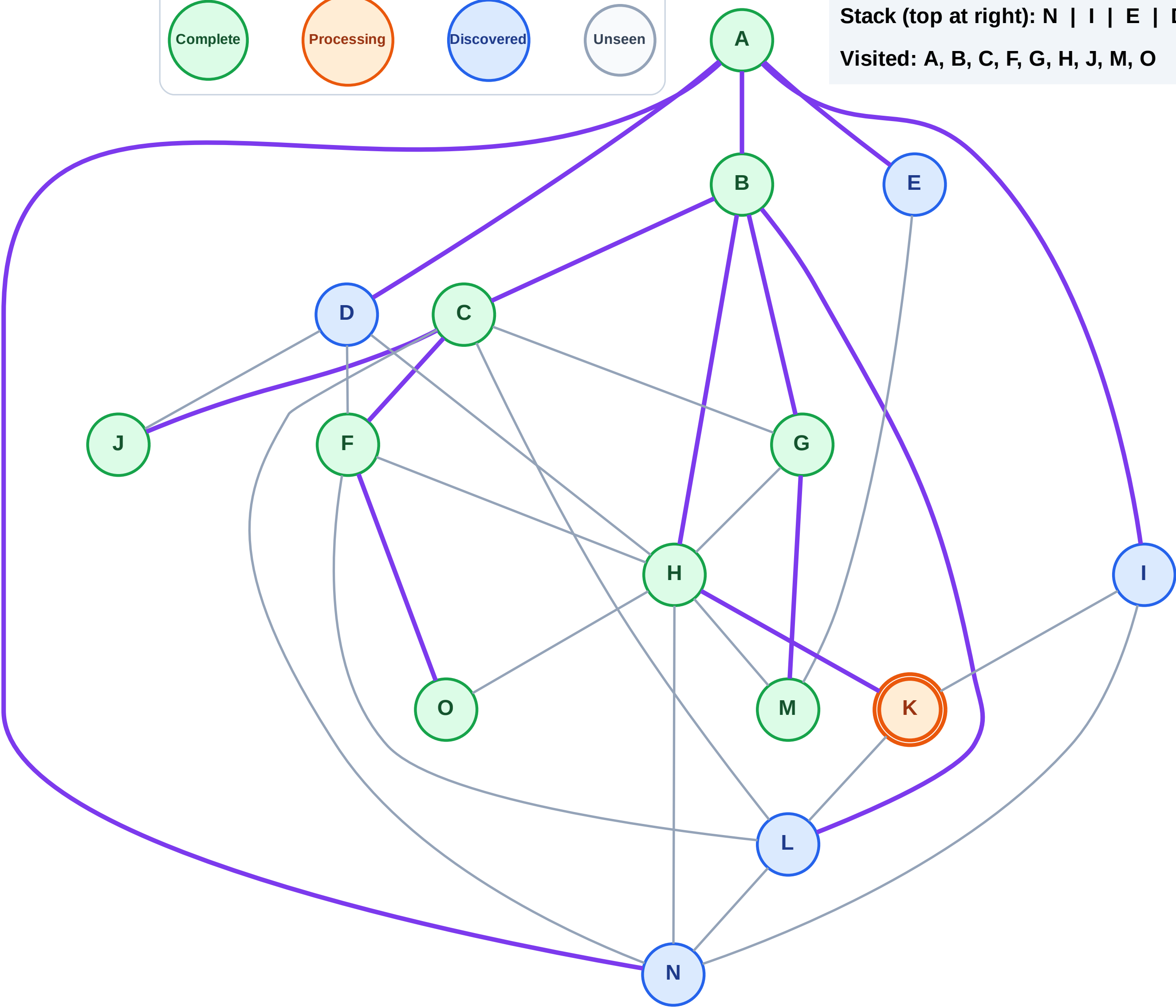
Step 20: Process node K

Legend



Stack (top at right): N | I | E | D | L

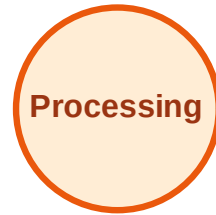
Visited: A, B, C, F, G, H, J, M, O



DFS Traversal

Step 21: TARGET NODE K REACHED! Path (3 hops): A → B → H → K

Legend



Stack (top at right): N | I | E | D | L

Visited: A, B, C, F, G, H, J, K, M, O

